New L J Institute of Engineering & Technology

SEM: 1st – CSE/CSE (AIML)/DS

Sub: Mathematics-I

Sub Code:BE01R00041

Module 3 : SEQUENCE AND SERIES

Weightage: 20%

SEQUENCE AND SERIES

TOPIC:1 SEQUENCES

Definition: A function whose domain is the set of positive integers is called a sequence.

- An ordered set of real numbers $u_1, u_2, u_3, \dots, u_n, \dots$ is called a sequence.
- It is denoted by $\{u_n\}$
- u_n is called the nth term of the sequence.

Limit of a Sequence

A sequence $\{u_n\}$ tends to a real number l as $n \to \infty$ if for every $\varepsilon > 0$ there exists a number m such that,

$|u_n - l| < \varepsilon \; for \; all \; n > m$ i.e. $\lim_{n\to\infty} u_n = l$

Continuous Function Theorem for Sequences

Let $\{u_n\}$ be a sequence of real numbers. If $u_n \to l$ and if f is a continuous function at l and defined at all u_n, then $f(u_n)u_n \to f(l)$

- **Convergent sequence:** A sequence which has a finite limit is called a convergent sequence.
- **Divergent sequence:** A sequence which has an infinite limit is called a divergent sequence.
- **Oscillatory sequence:** If the limit of the sequence is not unique, the sequence is said to be Oscillatory sequence.
- **Monotonic sequence:**

  A sequence is said to be **monotonically increasing** if

  $\boldsymbol{u_n \le u_{n+1}}$ **for each n**

  $\boldsymbol{e.g. 1, 2, 3, 4, 5 \dots}$

  A sequence is said to be **monotonically decreasing** if

  $\boldsymbol{u_{n+1} \le u_n}$ **for each n** $\quad e.g. \; \boldsymbol{1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots}$
- **Alternating Sequence:**

  A sequence is said to be alternating if the terms are alternatively positive and negative.

  e.g.1,-2, 3,-4…

- **Bounded Sequence:**

A sequence is said to be bounded if there exists numbers m and M such that $m < u_n < M, for\ all\ n$

NOTE:

(1)Every convergent sequence is bounded but the converse is not true.
E.g. $\{(-1)^n\}$ is bounded but not convergent

(2)A monotonic increasing sequence converges if it is bounded above,
And diverges to $+\infty$ if it is not bounded above.

(3) A monotonic decreasing sequence converges if it is bounded below
And diverges to $-\infty$ if it is not bounded below.

SANDWICH THEOREM:

Let $\{a_n\}, \{b_n\}$ and $\{c_n\}$ be three sequences such that $a_n \le b_n \le c_n$ for all n

$$If\ \lim_{n\to\infty} a_n = \lim_{n\to\infty} c_n = l\ then\ \lim_{n\to\infty} b_n = l$$

SHORT QUESTIONS

TYPE :1 TO CHECK MONOTONIC(INCREASING OR DECREASING), BOUNDEDNESS AND CONVERGENCE OF THE SEQUENCE

1 Does the sequence $\left\{\frac{3}{n+3}\right\}$ monotone?

Solution:

$$a_n = \frac{3}{n+3} \Rightarrow a_{n+1} = \frac{3}{(n+3)+1} = \frac{3}{n+4}$$

$$a_{n+1} - a_n = \frac{3}{n+4} - \frac{3}{n+3} = \frac{3(n+3-n-4)}{(n+4)(n+3)} = \frac{(-3)}{(n+4)(n+3)} < 0$$

$$\Longrightarrow a_{n+1} < a_n$$

Hence $\{a_n\}$ ***is monotonically decreasing.*** **Answer**

2 Check that the sequence $a_n = \frac{n}{n^2+1}$ is a decreasing and bounded below. Is it convergent? **OR**

Show that the sequence $\left\{\frac{n}{n^2+1}\right\}$is monotonic decreasing and bounded. Is it convergent?

Solution:

(1)To show:$\{a_n\}$ ***is monotonic decreasing***

$$a_n = \frac{n}{n^2+1} \Rightarrow a_{n+1} = \frac{n+1}{(n+1)^2+1} = \frac{n+1}{n^2+2n+2}$$

	$a_{n+1} - a_n = \frac{n+1}{n^2+2n+2} - \frac{n}{n^2+1} = \frac{(n+1)(n^2+1) - n(n^2+2n+2)}{\{n^2+2n+2\}(n^2+1)}$ $= \frac{n^3+n^2+n+1-n^3-2n^2-2n}{(n^2+2n+2)(n^2+1)} = \frac{-n^2-n+1}{(n^2+2n+2)(n^2+1)} < 0$ $= \frac{-n^2-n+1}{\{(n+1)^2+1\}(n^2+1)} < 0, \quad for\ all\ n \in N$ $\Longrightarrow a_{n+1} < a_n$ **Hence** $\{a_n\}$ ***is monotonic decreasing*** **(2) To show:** $\{a_n\}$ ***is bounded*** $a_n = \frac{n}{n^2+1}$ $a_1 = \frac{1}{2}, \ a_2 = \frac{2}{5}, \ a_3 = \frac{3}{10}, a_4 = \frac{4}{17}, \ldots$ **It is clear that** $a_n \leq \frac{1}{2}$, $for\ all\ n \in N$ $\therefore \{a_n\}$ ***is bounded below*** **Thus,** $\{a_n\}$ ***is decreasing and bounded below sequence.*** **Hence, it is convergent. Answer**
3	Show that the sequence $\{a_n\}$ whose n[th] term is $a_n = \frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + \cdots + \frac{1}{n!} n \in N$ is monotonic increasing and bounded. Is it convergent? **Solution:** $a_n = \frac{1}{1!} + \frac{1}{2!} + \cdots + \frac{1}{n!} \Rightarrow a_{n+1} = \frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} \ldots + \frac{1}{n!} + \frac{1}{(n+1)!}$ **(1)To show:** $\{a_n\}$ ***is monotonic increasing*** $a_{n+1} - a_n = \frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} \ldots + \frac{1}{n!} + \frac{1}{(n+1)!} - \left(\frac{1}{1!} + \frac{1}{2!} + \cdots + \frac{1}{n!}\right) = \frac{1}{(n+1)!} > 0$ $\Longrightarrow a_{n+1} > a_n$ **Hence** $\{a_n\}$ ***is monotonically increasing.*** **(2) To show:** $\{a_n\}$ ***is bounded***

	$a_n = \frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} \dots + \frac{1}{n!}$ $\leq \frac{1}{2^0} + \frac{1}{2^1} + \frac{1}{2^2} + \frac{1}{2^3} + \cdots + \frac{1}{2^{n-1}}$ $= \frac{1\left(1-\left(\frac{1}{2}\right)^n\right)}{1-\left(\frac{1}{2}\right)}$ $\left[\begin{array}{c} Since\ \frac{1}{2^0} + \frac{1}{2^1} + \frac{1}{2^2} + \frac{1}{2^3} + \cdots + \frac{1}{2^{n-1}} \text{ is in geometric progression,} \\ S_n = \frac{a(1-r^n)}{1-r};\ \text{here a = first term = 1, r = ratio} = \frac{second\ term}{first\ term} = \frac{1/2}{1} = 1/2 \end{array}\right]$ $= 2\left(1 - \frac{1}{2^n}\right)$ $= 2 - \frac{1}{2^{n-1}}$ < 2 $\therefore \{a_n\}\ is\ bounded$ **We know,** **A monotonic sequence is convergent⟺It is bounded** $\therefore \{a_n\}\ is\ convergent.$ **Answer**
4	Define the convergence of a sequence (a_n) and verify whether the sequence whose n^{th} term is $a_n = (\frac{n+1}{n-1})^n$ convergent or not. **Solution:** **Definition: Convergence of a Sequence** **A sequence $\{u_n\}$ is said to be convergent if the limit of the sequence is finite.** **i.e.** $\lim_{n\to\infty} u_n = l\ (finite)$ $\lim_{n\to\infty} a_n = \lim_{n\to\infty} \left(\frac{n+1}{n-1}\right)^n$ $= \frac{\lim_{n\to\infty} n^n \left(1+\frac{1}{n}\right)^n}{\lim_{n\to\infty} n^n \left[\left(1-\frac{1}{n}\right)^{-n}\right]^{(-1)}}$ $= \frac{e}{e^{(-1)}} = e^2, finite$ $\therefore \{a_n\}\ is\ convergent\ and\ \lim_{n\to\infty} a_n = e^2$ **Answer**
5	Test the convergence of sequence $\left(\frac{n^2+n}{2n^2-n}\right)$.

	Solution: $u_n = \left(\frac{n^2+n}{2n^2-n}\right)$ $\lim_{n\to\infty} u_n = \lim_{n\to\infty}\left(\frac{n^2+n}{2n^2-n}\right) = \lim_{n\to\infty}\frac{n^2}{n^2}\left(\frac{1+\frac{1}{n}}{2-\frac{1}{n}}\right) = \frac{1+0}{2-0} = \left(\frac{1}{2}\right), finite$ **Hence, the sequence converges** **Answer**
6	Find convergence of sequence $\left\{\frac{(-1)^n}{2}\right\}$. **Solution:** $u_n = \left\{\frac{(-1)^n}{2}\right\}$ $\lim_{n\to\infty} u_n = \lim_{n\to\infty}\left\{\frac{(-1)^n}{2}\right\}$ $= \frac{1}{2},\ if\ n\ is\ even$ $= \left(\frac{-1}{2}\right),\ if\ n\ is\ odd$ **Since limit is not unique, the sequence is Oscillatory.** **Answer**
7	Test the convergence of the sequence $\{2-(-1)^n a\}$ **Solution:** $u_n = \{2-(-1)^n a\}$ $\lim_{n\to\infty} u_n = \lim_{n\to\infty}\{2-(-1)^n a\}$ $= 2-a,\ if\ n\ is\ even$ $= 2+a,\ if\ n\ is\ odd$ **Since limit is not unique, the sequence is Oscillatory.** **Answer**
	TYPE:2 EXAMPLES BASED ON SANDWICH(SQUEEZING) THEOREM
8	State sandwich theorem for sequences and using it prove that the sequence $\left\{\frac{\sin n}{n}\right\}_{n=1}^{n=\infty}$ *is* convergent and $\lim_{n\to\infty}\frac{\sin n}{n} = 0$ **Solution:**

	Sandwich Theorem: Let $\{a_n\}, \{b_n\}$ *and* $\{c_n\}$ be sequences of real numbers such that $a_n \le b_n \le c_n\,,\ for\ all\ n$ If $\lim\limits_{n\to\infty} a_n = \lim\limits_{n\to\infty} b_n = l\,, then\ \lim\limits_{n\to\infty} c_n = l$ **To find:** $\lim\limits_{n\to\infty} \frac{\sin n}{n}$ **We know that,** $(-1) \le \sin n \le 1$ $\Rightarrow \left(\frac{-1}{n}\right) \le \frac{\sin n}{n} \le \frac{1}{n}$ $\lim\limits_{n\to\infty}\left(\frac{-1}{n}\right) = 0\ \ and\ \ \lim\limits_{n\to\infty}\left(\frac{1}{n}\right) = 0$ **By Sandwich theorem,** $: \lim\limits_{n\to\infty} \frac{\sin n}{n} = 0$ **, finite** **Hence, the sequence converges to zero.** **Answer**
9	Test the convergence of the sequence $\left\{\frac{\cos n}{n}\right\}$ **Solution:** **We know that,** $(-1) \le \cos n \le 1$ $\Rightarrow \left(\frac{-1}{n}\right) \le \frac{\cos n}{n} \le \frac{1}{n}$ $\lim\limits_{n\to\infty}\left(\frac{-1}{n}\right) = 0\ \ and\ \ \lim\limits_{n\to\infty}\left(\frac{1}{n}\right) = 0$ **By Sandwich theorem,** $: \lim\limits_{n\to\infty} \frac{\cos n}{n} = 0$ **, finite** **Hence, the sequence converges to zero.** **Answer**
10	State Sandwich theorem and using it find if $\lim\limits_{x\to 0} g(x)$ if $3 - x^2 \le g(x) \le 3\sec x$ for all x. **Solution:** **Sandwich Theorem:** Let $\{a_n\}, \{b_n\}$ *and* $\{c_n\}$ be sequences of real numbers such that

	$a_n \le b_n \le c_n$, $for\ all\ n$ If $\lim_{n\to\infty} a_n = \lim_{n\to\infty} b_n = l\ , then\ \lim_{n\to\infty} c_n = l$ **To find :** $\lim_{x\to 0} g(x)$ $3 - x^2 \le g(x) \le 3\sec x$ for all x $\lim_{x\to 0} 3 - x^2 = 3 - 0 = 3\ \ and\ \ \lim_{x\to 0} 3\sec x = 3\sec 0 = 3$ **By Sandwich theorem,** $\lim_{x\to 0} g(x) = 3$
11	If $\frac{1}{2} - \frac{x^2}{24} < \frac{1-\cos x}{x^2} < \frac{1}{2}$ hold for all values of x close to zero. Find $\lim_{x\to 0} \frac{1-\cos x}{x^2}$ **Solution:** **To find :** $\lim_{x\to 0} \frac{1-\cos x}{x^2}$ $\frac{1}{2} - \frac{x^2}{24} < \frac{1-\cos x}{x^2} < \frac{1}{2}$ $\lim_{x\to 0}\left(\frac{1}{2} - \frac{x^2}{24}\right) = \frac{1}{2} - 0 = \frac{1}{2}\ \ and\ \lim_{x\to 0}\left(\frac{1}{2}\right) = \frac{1}{2}$ **By Sandwich theorem,** $\lim_{x\to 0} \frac{1-\cos x}{x^2} = \frac{1}{2}$
12	If $\sqrt{5-2x^2} \le f(x) \le \sqrt{5-x^2}$ for $-1 \le x \le 1$ find $\lim_{x\to 0} f(x)$ using Sandwich theorem. **Solution: To find :** $\lim_{x\to 0} f(x)$ $\sqrt{5-2x^2} \le f(x) \le \sqrt{5-x^2}$ $\lim_{x\to 0} \sqrt{5-2x^2} = \sqrt{5-0} = \sqrt{5}\ \ and\ \ \lim_{x\to 0} \sqrt{5-x^2} = \sqrt{5-0} = \sqrt{5}$ **By Sandwich theorem,** $\lim_{x\to 0} f(x) = \sqrt{5}$ **Answer**
13	State sandwich theorem on sequences and using it show that, if x = R with $\lvert x\rvert < 1$, then $x^n \to 0\ as\ n \to \infty$ **Solution:**

Given: x = R with $|x| < 1$

To show: $x^n \to 0 \; as \; n \to \infty$

Case :1 For x = 0, $x^n = 0$

Case :2 For $x \neq 0$,

We have $|x| < 1$

$$\Rightarrow \frac{1}{|x|} > 1$$

$$\Rightarrow \frac{1}{|x|} = 1 + h \; where \; h > 0$$

$$\Rightarrow \left(\frac{1}{|x|}\right)^n = (1+h)^n$$

$$\Rightarrow \frac{1}{|x|^n} = 1 + nh + \frac{n(n-1)}{2!}h^2 + \cdots (Using \; Binomial \; Expansion)$$

$$> nh$$

$$\Rightarrow |x|^n < \left(\frac{1}{nh}\right)$$

$$Thus, 0 < |x|^n < \left(\frac{1}{nh}\right)$$

$$\lim_{n\to\infty} 0 = 0 \quad and \quad \lim_{n\to\infty}\left(\frac{1}{nh}\right) = 0$$

By Sandwich Theorem,

$$\lim_{n\to\infty}|x|^n = 0 \qquad \text{Hence}, x^n \to 0 \; as \; n \to \infty$$ **Answer**

TOPIC:2 CONVERGENCE AND DIVERGENCE OF AN INFINITE SERIES

Infinite Series: Let $u_1, u_2, u_3, u_4, \ldots, u_n, \ldots$ be an infinite sequence of real numbers, then an expression of the form $u_1 + u_2 + u_3 + \cdots + u_n + \cdots$is called an infinite series.

It is denoted by $\sum_{n=1}^{\infty} u_n$ or $\sum u_n$

Thus, $\sum_{n=1}^{\infty} u_n = u_1 + u_2 + u_3 + \cdots + u_n + \cdots$

The number u_n is known as n^{th} term of the series.

$S_n = The \; sum \; of \; first \; n \; terms \; of \; the \; series$

$= n^{th}$ partial sum

$= u_1 + u_2 + u_3 + \cdots + u_n$

It is not necessary that partial sum always increase, it depends on the sign of the terms of u_n

$\boldsymbol{S_1 = u_1}$

$\boldsymbol{S_2 = u_1 + u_2}$

$\boldsymbol{S_3 = u_1 + u_2 + u_3}$

And so on.

$\boldsymbol{S_n = u_1 + u_2 + u_3 + \cdots + u_n}$

Convergence, Divergence and oscillation of an Infinite Series

Convergence Criteria

- A series $\sum u_n$ is said to be convergent if $\lim_{n\to\infty} S_n = S \ (finite)$
- A series $\sum u_n$ is said to be divergent if $\lim_{n\to\infty} S_n \to \pm\infty$
- A series $\sum u_n$ is said to be Oscillatory if limit is not unique.

TOPIC:3 GEOMETRIC SERIES

Definition:

An infinite series of the form $a + ar + ar^2 + ar^3 + \cdots$ is known as geometric series.

This series is

(1)Convergent, if $|r| < 1$ $(i.e. (-1) < r < 1)$ and its sum is $S = \frac{a}{1-r}$

(2)Divergent, if $r \geq 1$

(3)Finitely Oscillating if $r = (-1)$

	(4)Infinitely Oscillating if $r < (-1)$ **NOTE:** If the series $\sum u_n$ *and* $\sum v_n$ converges to l and m, respectively, then $\sum(u_n \pm v_n)$ is also convergent and converges to $(l \pm m)$.
	SHORT QUESTIONS
1	The sum of the series $\sum_{n=1}^{\infty}\left(\frac{e}{\pi}\right)^n$ is **Solution:** $\sum_{n=1}^{\infty}\left(\frac{e}{\pi}\right)^n = \left(\frac{e}{\pi}\right)^1 + \left(\frac{e}{\pi}\right)^2 + \left(\frac{e}{\pi}\right)^3 + \cdots$ *is a geometric seies with* $a = \frac{e}{\pi}, r = \frac{e}{\pi} < 1$ Sum $S = \frac{a}{1-r} = \frac{\frac{e}{\pi}}{1-\frac{e}{\pi}} = \frac{e}{\pi - e}$
2	The sum of the series $2 + \frac{2}{3} + \frac{2}{9} + \frac{2}{27} \ldots$ *is* ____ **Solution:** $2 + \frac{2}{3} + \frac{2}{9} + \frac{2}{27} \ldots$ *is* a geometric series with , $a = 2,\ r = \frac{2/3}{2} = \frac{1}{3} < 1$ $\therefore$ The series is convergent and sum is $S = \frac{a}{1-r} = \frac{2}{1-1/3} = 3$
3	Sum of the series $\sum_{n=0}^{\infty}\frac{4}{2^n}$ is__ **Solution:** $\sum_{n=0}^{\infty}\frac{4}{2^n} = 4 + \frac{4}{2^1} + \frac{4}{2^2} + \cdots$is a geometric series with , $a = 4,\ r = \frac{2}{4} = \frac{1}{2} < 1$ $\therefore$ The series is convergent and sum is $S = \frac{a}{1-r} = \frac{4}{1-1/2} = 8$
4	The sum of the series $1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} \ldots$ *is* ___ **Solution:** $1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} \ldots$ is a geometric series with , $a = 1, r = \frac{1}{2} < 1$ $\therefore$ The series is convergent and sum is $S = \frac{a}{1-r} = \frac{1}{1-1/2} = 2$
5	Test the convergence of $\sum_{n=0}^{\infty}\frac{2^n}{3^n}$ **Solution:**

	$\sum_{n=0}^{\infty} \frac{2^n}{3^n} = \sum_{n=1}^{\infty} \left(\frac{2}{3}\right)^n = \left(\frac{2}{3}\right)^1 + \left(\frac{2}{3}\right)^2 + \left(\frac{2}{3}\right)^3 + \cdots \text{ is a geometric seies}$ $\text{with } a = \frac{2}{3}, r = \frac{2}{3} < 1$ ∴ The series is convergent
6	Define geometric series and show that the geometric series is (i) convergent if |r| <1 (ii) divergent if r ≥ 1 **Solution:** **Definition:** An infinite series of the form $a + ar + ar^2 + ar^3 + \cdots + ar^{n-1} + ar^n + \cdots$ is known as **geometric series.** Consider the geometric series $a + ar + ar^2 + ar^3 + \cdots + ar^{n-1} + ar^n + \cdots$ $S_n = a + ar + ar^2 + ar^3 + \cdots + ar^{n-1}$ $= \frac{a(1-r^n)}{1-r}$, if $r < 1$ $= \frac{a(r^n-1)}{r-1}$, if $r > 1$ (1) When $\|r\| < 1$, $\lim_{n\to\infty} r^n = 0$ $\lim_{n\to\infty} S_n = \frac{a}{1-r}$ is finite Hence, the series is convergent. (2) When $r > 1$, $\lim_{n\to\infty} r^n \to \infty$ $\lim_{n\to\infty} S_n = \lim_{n\to\infty} \frac{a(r^n-1)}{r-1} \to \infty$ Hence, the series is divergent. (3) When $r = 1$, $S_n = a + a + a + a + \cdots + a = na$ $\lim_{n\to\infty} S_n \to \infty$ Hence, the series is divergent.
	NUMERICAL QUESTIONS

1	Determine whether the following series converge or diverge. Find the sum of the series if converges $5-\frac{10}{3}+\frac{20}{9}-\frac{40}{27}+\cdots$ **Solution:** The given series is a geometric series with $a=5\ and\ r=\frac{\left(\frac{-10}{3}\right)}{5}=\left(\frac{-2}{3}\right)$ $\lvert r\rvert=\left\lvert\left(\frac{-2}{3}\right)\right\rvert=\frac{2}{3}<1$ Hence, the series is convergent. Sum of the series is $S=\frac{a}{1-r}=\frac{5}{1-\left(-\frac{2}{3}\right)}=\frac{5}{1+\frac{2}{3}}=3$ **Answer**
2	Find the sum of following series if it converges: $1+\frac{1}{3}+\frac{1}{9}+\frac{1}{27}+\cdots$ **Solution:** The given series is a geometric series with $a=1\ and\ r=\frac{1}{3}$ $\lvert r\rvert=\frac{1}{3}<1$ Hence, the series is convergent. Sum of the series is $S=\frac{a}{1-r}=\frac{1}{1-\left(\frac{1}{3}\right)}=\frac{3}{2}$ **Answer**
3	Prove that $1+\frac{2}{3}+\frac{4}{9}+\frac{8}{27}+\frac{16}{81}+\ldots..$ converges and find it's sum. **Solution:** The given series is a geometric series with $a=1\ and\ r=\frac{2}{3}$ $\lvert r\rvert=\frac{2}{3}<1$ Hence, the series is convergent. Sum of the series is

	$S = \frac{a}{1-r} = \frac{1}{1-\left(\frac{2}{3}\right)} = 3$ **Answer**
4	Test the convergence of $\sum_{n=0}^{\infty} \frac{3^{2n}}{2^{3n}}$ **Solution:** $\sum_{n=0}^{\infty} u_n = \sum_{n=0}^{\infty} \frac{3^{2n}}{2^{3n}} = \sum_{n=0}^{\infty} \left(\frac{3^2}{2^3}\right)^n = \sum_{n=0}^{\infty} \left(\frac{9}{8}\right)^n = 1 + \left(\frac{9}{8}\right) + \left(\frac{9}{8}\right)^2 + \left(\frac{9}{8}\right)^3 + \cdots$ This is a geometric series with $a = 1$ and $r = \frac{9}{8} > 1$ Hence, the series is divergent. **Answer**
5	Find the sum of the series $\sum_{n\geq 2} \frac{1}{4^n}$ **Solution:** $\sum_{n\geq 2} u_n = \sum_{n\geq 2} \frac{1}{4^n} = \frac{1}{4^2} + \frac{1}{4^3} + \frac{1}{4^4} + \frac{1}{4^5} + \cdots$ This is a geometric series with $a = \frac{1}{4^2}$ and $r = \frac{\frac{1}{4^3}}{\frac{1}{4^2}} = \frac{1}{4} < 1$ Hence, the series is convergent. Sum of the series is $S = \frac{a}{1-r} = \frac{\frac{1}{4^2}}{1-\left(\frac{1}{4}\right)} = \frac{4}{3\times 16} = \frac{1}{12}$ **Answer**
6	Test the convergence and Divergence of the following series $\sum_{n=1}^{\infty} \frac{4^n + 5^n}{6^n}$ **Solution:** $\sum_{n=1}^{\infty} u_n = \sum_{n=1}^{\infty} \frac{4^n + 5^n}{6^n} = \sum_{n=1}^{\infty} \left(\frac{4}{6}\right)^n + \sum_{n=1}^{\infty} \left(\frac{5}{6}\right)^n = \sum_{n=1}^{\infty} a_n + \sum_{n=1}^{\infty} b_n$ $\sum_{n=1}^{\infty} a_n = \sum_{n=1}^{\infty} \left(\frac{4}{6}\right)^n = \sum_{n=1}^{\infty} \left(\frac{2}{3}\right)^n = \frac{2}{3} + \left(\frac{2}{3}\right)^2 + \left(\frac{2}{3}\right)^3 + \left(\frac{2}{3}\right)^4 + \cdots$ This is a geometric series with $a = \frac{2}{3}$ and $r = \frac{\left(\frac{2}{3}\right)^2}{\frac{2}{3}} = \frac{2}{3} < 1$ Hence, the series is convergent.

	$\sum_{n=1}^{\infty} b_n = \sum_{n=1}^{\infty} \left(\frac{5}{6}\right)^n = \frac{5}{6} + \left(\frac{5}{6}\right)^2 + \left(\frac{5}{6}\right)^3 + \left(\frac{5}{6}\right)^4 + \cdots$ This is a geometric series with $a = \frac{5}{6}$ and $r = \frac{\left(\frac{5}{6}\right)^2}{\frac{5}{6}} = \frac{5}{6} < 1$ Hence, the series is convergent. $\therefore \sum_{n=1}^{\infty} u_n = \sum_{n=1}^{\infty} a_n + \sum_{n=1}^{\infty} b_n$ is also convergent.
7	Test the convergence of the series $\sum_{n=0}^{\infty}\left(\frac{1}{2^n} + \frac{1}{3^n}\right)$ **Solution:** $\sum_{n=0}^{\infty} u_n = \sum_{n=0}^{\infty}\left(\frac{1}{2^n} + \frac{1}{3^n}\right) = \sum_{n=0}^{\infty}\left(\frac{1}{2}\right)^n + \sum_{n=0}^{\infty}\left(\frac{1}{3}\right)^n = \sum_{n=0}^{\infty} a_n + \sum_{n=0}^{\infty} b_n$ $\sum_{n=0}^{\infty} a_n = \sum_{n=0}^{\infty}\left(\frac{1}{2}\right)^n = 1 + \left(\frac{1}{2}\right)^1 + \left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^3 + \cdots$ This is a geometric series with $a = 1$ and $r = \frac{1}{2} < 1$ Hence, the series is convergent. $\sum_{n=0}^{\infty} b_n = \sum_{n=0}^{\infty}\left(\frac{1}{3}\right)^n = 1 + \left(\frac{1}{3}\right)^1 + \left(\frac{1}{3}\right)^2 + \left(\frac{1}{3}\right)^3 + \cdots$ This is a geometric series with $a = 1$ and $r = \frac{1}{3} < 1$ Hence, the series is convergent. $\therefore \sum_{n=0}^{\infty} u_n = \sum_{n=0}^{\infty} a_n + \sum_{n=0}^{\infty} b_n$ is also convergent.
8	Test the convergence of the series. If converges find the sum. $\sum_{n=0}^{n=\infty} \frac{1+2^n}{5^n}$ **Solution:** $\sum_{n=0}^{\infty} u_n = \sum_{n=0}^{\infty}\left(\frac{1+2^n}{5^n}\right) = \sum_{n=0}^{\infty}\left(\frac{1}{5}\right)^n + \sum_{n=0}^{\infty}\left(\frac{2}{5}\right)^n = \sum_{n=0}^{\infty} a_n + \sum_{n=0}^{\infty} b_n$ $\sum_{n=0}^{\infty} a_n = \sum_{n=0}^{\infty}\left(\frac{1}{5}\right)^n = 1 + \left(\frac{1}{5}\right)^1 + \left(\frac{1}{5}\right)^2 + \left(\frac{1}{5}\right)^3 + \cdots$

Exploring Emerging Technologies

	This is a geometric series with $a = 1$ and $r = \frac{1}{5} < 1$ Hence, the series is convergent. $\sum_{n=0}^{\infty} b_n = \sum_{n=0}^{\infty} \left(\frac{2}{5}\right)^n = 1 + \left(\frac{2}{5}\right)^1 + \left(\frac{2}{5}\right)^2 + \left(\frac{2}{5}\right)^3 + \cdots$ This is a geometric series with $a = 1$ and $r = \frac{2}{5} < 1$ Hence, the series is convergent. $\therefore \sum_{n=0}^{\infty} u_n = \sum_{n=0}^{\infty} a_n + \sum_{n=0}^{\infty} b_n$ is also convergent
9	Test the convergence or Divergence of the following series $\sum_{n=1}^{\infty} \frac{2^n - 1}{3^n}$ **Solution:** $\sum_{n=1}^{\infty} u_n = \sum_{n=1}^{\infty} \frac{2^n - 1}{3^n} = \sum_{n=1}^{\infty} \left(\frac{2}{3}\right)^n - \sum_{n=1}^{\infty} \left(\frac{1}{3}\right)^n = \sum_{n=1}^{\infty} a_n - \sum_{n=1}^{\infty} b_n$ $\sum_{n=1}^{\infty} a_n = \sum_{n=1}^{\infty} \left(\frac{2}{3}\right)^n = \left(\frac{2}{3}\right)^1 + \left(\frac{2}{3}\right)^2 + \left(\frac{2}{3}\right)^3 + \cdots$ This is a geometric series with $a = \frac{2}{3}$ and $r = \frac{2}{3} < 1$ Hence, the series is convergent. $\sum_{n=1}^{\infty} b_n = \sum_{n=1}^{\infty} \left(\frac{1}{3}\right)^n = \left(\frac{1}{3}\right)^1 + \left(\frac{1}{3}\right)^2 + \left(\frac{1}{3}\right)^3 + \left(\frac{1}{3}\right)^4 + \cdots$ This is a geometric series with $a = \frac{1}{3}$ and $r = \frac{\left(\frac{1}{3}\right)^2}{\frac{1}{3}} = \frac{1}{3} < 1$ Hence, the series is convergent. $\therefore \sum_{n=1}^{\infty} u_n = \sum_{n=1}^{\infty} a_n - \sum_{n=1}^{\infty} b_n$ is also convergent.
10	Define the Geometric series and find the sum of the following series $\sum_{n=1}^{\infty} \frac{3^{n-1} - 1}{6^{n-1}}$ **Solution:** $\sum_{n=1}^{\infty} u_n = \sum_{n=1}^{\infty} \frac{3^{n-1} - 1}{6^{n-1}} = \sum_{n=1}^{\infty} \left(\frac{3}{6}\right)^{n-1} - \sum_{n=1}^{\infty} \left(\frac{1}{6}\right)^{n-1} = \sum_{n=1}^{\infty} a_n - \sum_{n=1}^{\infty} b_n$ $\sum_{n=1}^{\infty} a_n = \sum_{n=1}^{\infty} \left(\frac{1}{2}\right)^{n-1} = \left(\frac{1}{2}\right)^0 + \left(\frac{1}{2}\right)^1 + \left(\frac{1}{2}\right)^2 + \cdots$

This is a geometric series with $a = 1 \; and \; r = \frac{1}{2} < 1$

Hence, the series is convergent.

Sum of the series is

$$S_1 = \frac{a}{1-r} = \frac{1}{1-\left(\frac{1}{2}\right)} = 2$$

$$\sum_{n=1}^{\infty} b_n = \sum_{n=1}^{\infty} \left(\frac{1}{6}\right)^{n-1} = \left(\frac{1}{6}\right)^0 + \left(\frac{1}{6}\right)^1 + \left(\frac{1}{6}\right)^2 + \left(\frac{1}{6}\right)^3 + \cdots$$

This is a geometric series with $a = 1 \; and \; r = \frac{1}{6} < 1$

Hence, the series is convergent.

Sum of the series is

$$S_2 = \frac{a}{1-r} = \frac{1}{1-\left(\frac{1}{6}\right)} = \frac{6}{5}$$

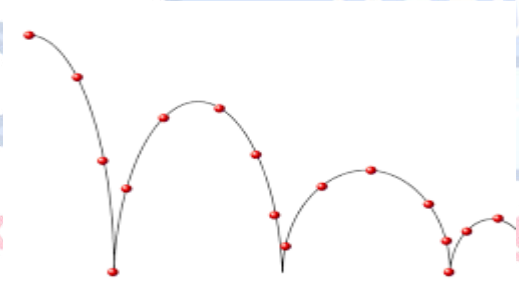
Sum $\sum_{n=1}^{\infty} \frac{3^{n-1}-1}{6^{n-1}}$

$\therefore \sum_{n=1}^{\infty} u_n = \sum_{n=1}^{\infty} a_n - \sum_{n=1}^{\infty} b_n$ is also convergent.

Sum of the series $\sum_{n=1}^{\infty} \frac{3^{n-1}-1}{6^{n-1}} = \sum_{n=1}^{\infty} \left(\frac{3}{6}\right)^{n-1} - \sum_{n=1}^{\infty} \left(\frac{1}{6}\right)^{n-1} \; is \; S_1 - S_2 = 2 - \frac{6}{5} = \frac{4}{5}$

11 You drop a ball from a meter above a flat surface. Each time the ball hits the surface after falling a distance h, it rebounds a distance *rh*, where $0 < r < 1$. Find the total distance ball travels up and down where $a = 6\ m$ and $r = \frac{2}{3}$ m

Solution:



BOUNCING BALL

The total distance travels by the ball is

$= a + 2ar + 2ar^2 + 2ar^3 + \cdots$

$= a + 2ar(1 + r + r^2 + r^3 + \cdots)$

$= a + 2ar\left(\frac{1}{1-r}\right)$

$(\because 1 + r + r^2 + \cdots \text{ is a geometric series with } a = 1, \text{ratio} = r, \text{sum } S = \frac{a}{1-r})$

$= \frac{a - ar + 2ar}{1-r}$

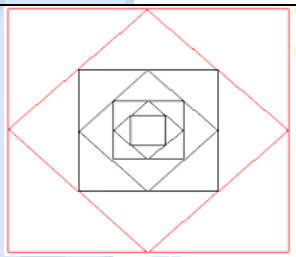
$= \frac{a + ar}{1-r}$

$= \frac{a(1+r)}{1-r}$

Here $a = 6\ m$ and $r = \frac{2}{3}\ m$.

$\therefore$ The total distance travels by the ball $= \frac{a(1+r)}{1-r} = \frac{6\left(1+\frac{2}{3}\right)}{1-\frac{2}{3}} = \frac{6\times\frac{5}{3}}{\frac{1}{3}} = 30\ m$

12



The figure shows that the first seven of a sequence of squares. The outermost square has an area of $4m^2$. Each of the other squares is obtained by joining the midpoints of the sides of the squares before it. Find the sum of the areas of all squares in the infinite sequence.

Solution:

Area of outermost square $= 4m^2$

Area of 2nd outermost square $= \frac{1}{2} \times 4 = 2m^2$

Area of 3rd outermost square $= \frac{1}{2} \times 2 = 1m^2$

Area of 4th outermost square $= \frac{1}{2} \times 1 = \frac{1}{2}m^2$ and so on.

Sum of Areas of all squares in the infinite sequence is

$S = 4 + 2 + 1 + \frac{1}{2} + \cdots$

$\text{which is an infinite geometric series with } a = 4 \text{ and } r = \frac{2}{4} = \frac{1}{2} < 1$

	$\therefore Sum\ S = \frac{a}{1-r} = \frac{4}{1-\frac{1}{2}} = 8m^2$
13	Let S=$\sum_{n=1}^{\infty} n\alpha^n$ where |α|<1. Find the value of α in (0,1) such that S=2α. **Solution:** $S = \sum_{n=1}^{\infty} n\alpha^n = 1\alpha + 2\alpha^2 + 3\alpha^3 + 4\alpha^4 + \cdots \quad (1)$ $\alpha S = \alpha^2 + 2\alpha^3 + 3\alpha^4 + 4\alpha^5 + \cdots \quad (2)$ Subtracting Eq. (2) from Eq. (1), $S(1-\alpha) = \alpha + \alpha^2 + \alpha^3 + \alpha^4 + \cdots$ $which\ is\ an\ infinite\ geometric\ series\ with\ a = \alpha\ \ and\ \ r = \frac{\alpha^2}{\alpha} = \alpha$ $\therefore\ S(1-\alpha) = \frac{\alpha}{1-\alpha}\ \left(As\ Sum = \frac{a}{1-r}\right)$ $\therefore S = \frac{\alpha}{(1-\alpha)^2}$ $If\ S = 2\alpha(given), then$ $2\alpha = \frac{\alpha}{(1-\alpha)^2}$ $\Rightarrow (1-\alpha)^2 = \frac{1}{2}$ $\Rightarrow 1-\alpha = \frac{1}{\sqrt{2}}\ \left(As\ \alpha \in (0,1), \quad 1-\alpha = \frac{-1}{\sqrt{2}}\ not\ possible\right)$ $\Rightarrow \alpha = 1 - \frac{1}{\sqrt{2}} = \frac{\sqrt{2}-1}{\sqrt{2}} = \frac{1.414-1}{1.414} = 0.2927$
14	Find the value of b for which $1 + e^b + e^{2b} + e^{3b} + \ldots + e^{nb} + \ldots = 9$ **Solution:** $\mathbf{1 + e^b + e^{2b} + e^{3b} + \cdots + e^{nb} + \cdots}\ is\ an\ infinite\ geometric\ series$ $with\ a = 1\ and\ r = \boldsymbol{e^b} < 1$ $\therefore Sum\ S = \frac{a}{1-r} = \frac{1}{1-\boldsymbol{e^b}}$ $\mathbf{1 + e^b + e^{2b} + e^{3b} + \cdots + e^{nb} + \cdots = 9}$

Exploring Emerging Technologies

	$\Rightarrow \frac{1}{1-e^b} = 9$ $\Rightarrow 1 - e^b = \frac{1}{9}$ $\Rightarrow e^b = 1 - \frac{1}{9} = \frac{8}{9}$ $\Rightarrow b \log_e e = log_e \left(\frac{8}{9}\right)$ $\Rightarrow b = \ln\left(\frac{8}{9}\right)$ $(\because log_e e = 1)$
	TOPIC:4 TELESCOPING SERIES
	NUMERICAL QUESTIONS
1	Determine whether the following series converges or diverges. Find the sum of the series if converges. $\sum_{n=1}^{\infty}[\tan^{-1}(n) - \tan^{-1}(n+1)]$ **Solution:** $\sum_{n=1}^{\infty} u_n = \sum_{n=1}^{\infty}[\tan^{-1}(n) - \tan^{-1}(n+1)]$ $= (\tan^{-1}1 - \tan^{-1}2) + (\tan^{-1}2 - \tan^{-1}3) + (\tan^{-1}3 - \tan^{-1}4) + \cdots + (\tan^{-1}(n) - \tan^{-1}(n+1)) + \cdots$ $S_n = \tan^{-1}1 - \tan^{-1}(n+1)$ $\lim_{n\to\infty} S_n = \lim_{n\to\infty}[\tan^{-1}1 - \tan^{-1}(n+1)] = \left[\frac{\pi}{4} - \tan^{-1}\infty\right] = \frac{\pi}{4} - \frac{\pi}{2} = \left(-\frac{\pi}{4}\right)$, finite Thus, the series is convergent and sum $= \left(-\frac{\pi}{4}\right)$
2	Test the convergence of the series $\sum_{n=1}^{\infty}\tan^{-1}\left(\frac{1}{n^2+n+1}\right)$ and if it is convergent then find its sum. **Solution:** $\sum_{n=1}^{\infty} u_n = \sum_{n=1}^{\infty}\tan^{-1}\left(\frac{1}{n^2+n+1}\right) = \sum_{n=1}^{\infty}\tan^{-1}\left(\frac{(n+1)-n}{1+n(n+1)}\right)$ $= \sum_{n=1}^{\infty}\tan^{-1}\left(\frac{\frac{n+1}{n(n+1)} - \frac{n}{n(n+1)}}{\frac{1}{n(n+1)} + \frac{n(n+1)}{n(n+1)}}\right) = \sum_{n=1}^{\infty}\tan^{-1}\left(\frac{\left(\frac{1}{n}\right) - \left(\frac{1}{n+1}\right)}{1 + \left(\frac{1}{n}\right)\left(\frac{1}{n+1}\right)}\right)$

Exploring Emerging Technologies

	$= \sum_{n=1}^{\infty}\left[\tan^{-1}\left(\frac{1}{n}\right) - \tan^{-1}\left(\frac{1}{n+1}\right)\right]$ $= \left[\tan^{-1}1 - \tan^{-1}\left(\frac{1}{2}\right)\right] + \left[\tan^{-1}\left(\frac{1}{2}\right) - \tan^{-1}\left(\frac{1}{3}\right)\right] + \left[\tan^{-1}\left(\frac{1}{3}\right) - \tan^{-1}\left(\frac{1}{4}\right)\right] + \cdots$ $+ \left[\tan^{-1}\left(\frac{1}{n}\right) - \tan^{-1}\left(\frac{1}{n+1}\right)\right] + \cdots$ $S_n = \left[\tan^{-1}1 - \tan^{-1}\left(\frac{1}{n+1}\right)\right]$ $\lim_{n\to\infty} S_n = \lim_{n\to\infty}\left[\tan^{-1}1 - \tan^{-1}\left(\frac{1}{n+1}\right)\right] = \lim_{n\to\infty}\left[\frac{\pi}{4} - \tan^{-1}\left(\frac{1}{n+1}\right)\right] = \frac{\pi}{4} - 0 = \left(\frac{\pi}{4}\right)$, finite Thus, the series is convergent and sum $= \left(\frac{\pi}{4}\right)$
3	Test the convergence of the series $\sum_{n=1}^{\infty}\frac{1}{n(n+1)}$, if convergent then find its value. OR Test for convergence the series $\frac{1}{1.2} + \frac{1}{2.3} + \frac{1}{3.4} + \cdots$ **Solution:** $\sum_{n=1}^{\infty} u_n = \sum_{n=1}^{\infty}\frac{1}{n(n+1)} = \sum_{n=1}^{\infty}\left(\frac{1}{n} - \frac{1}{n+1}\right)$ $= \left(\frac{1}{1} - \frac{1}{2}\right) + \left(\frac{1}{2} - \frac{1}{3}\right) + \left(\frac{1}{3} - \frac{1}{4}\right) + \left(\frac{1}{4} - \frac{1}{5}\right) + \cdots + \left(\frac{1}{n} - \frac{1}{n+1}\right) + \cdots$ $S_n = 1 - \frac{1}{n+1}$ $\lim_{n\to\infty} S_n = \lim_{n\to\infty}\left(1 - \frac{1}{n+1}\right) = (1-0) = 1, finite$ Thus, the series is convergent and sum = 1
	TOPIC:5 **The nth TERM TEST (ZERO TEST /DIVERGENCE TEST), COMBINING SERIES, HARMONIC SERIES**
	Positive term series An infinite series in which all terms, after some particular term, are positive is called a positive term series. E.g. -7-5-1+5+13+23+… is positive term series as all terms after third term are positive.

The series of positive terms is either convergent or divergent but cannot be oscillatory.

Necessary Condition for the Convergence of the Positive term series

If series $\sum u_n$ of positive terms is convergent, then $\lim_{n\to\infty} u_n = 0$

But the converse of the theorem is not true; that is

$\lim_{n\to\infty} u_n = 0$, then the series $\sum u_n$ may or may not converge.

Thus, $\lim_{n\to\infty} u_n = 0$ is necessary but not sufficient condition for the series $\sum u_n$ to be convergent.

For Example,

Consider the Harmonic series

$$\mathbf{1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \cdots + \frac{1}{n} + \cdots}$$

Here, $u_n = \frac{1}{n}$ and $\lim_{n\to\infty} u_n = \lim_{n\to\infty}\left(\frac{1}{n}\right) = 0$

But the series is known to be divergent.

The nth Term Test (Zero Test / Divergence Test)

If $\lim_{n\to\infty} u_n \neq 0$ or $\lim_{n\to\infty} u_n$ does not exist

Then $\sum u_n$ diverges.

Combination of two series

(1) If $\sum u_n$ and $\sum v_n$ are two convergent series then

- $\sum(u_n \pm v_n) = \sum u_n \pm \sum v_n$ is also convergent.
- $\sum(ku_n) = k\sum u_n,\ k\ is\ any\ number$ is also convergent .
- But $(\sum u_n)(\sum v_n)$may or may not be convergent.

(2) If $\sum u_n$ is convergent and $\sum v_n$ is divergent series then

- $\sum(u_n \pm v_n) = \sum u_n \pm \sum v_n$ is also divergent.
- If $\sum u_n$ is divergent series then $\sum(ku_n) = k\sum u_n$ is also divergent..

(3) If $\sum u_n$ and $\sum v_n$ are two divergent series then $\sum(u_n \pm v_n)$ may or may not be divergent.

E.g. let $\sum u_n = 2 + 2 + 2 + \cdots$ and $\sum v_n = (-2) + (-2) + (-2) + \cdots$

Then both $\sum u_n$ and $\sum v_n$ are two divergent series, whereas

$\sum(u_n + v_n) = 0 + 0 + 0 + \cdots$Converges to 0.

	NUMERICAL QUESTIONS
1	Does the series $\sum_{n=1}^{\infty}(1-e^{-n})$ converge? Justify. **Solution:** $\boldsymbol{u_n} = (1-e^{-n})$ $\lim_{n\to\infty} \boldsymbol{u_n} = \lim_{n\to\infty}(1-e^{-n}) = 1-e^{-\infty} = 1-\frac{1}{e^{\infty}} = 1-0 = 1 \neq 0\ (As\ e^{\infty} \to \infty$ Thus, by the nth term test (zero test) the series is divergent. **Answer**
2	Prove that the series $\sum_{n=1}^{\infty}\frac{n^2-1}{n^2+1}$ is divergent **Solution:** $\boldsymbol{u_n} = \frac{n^2-1}{n^2+1}$ $\lim_{n\to\infty} \boldsymbol{u_n} = \lim_{n\to\infty}\frac{n^2-1}{n^2+1} = \lim_{n\to\infty}\left(\frac{1-(1/n^2)}{1+(1/n^2)}\right) = 1 \neq 0$ Thus, by the nth term test (zero test) the series is divergent. **Answer**
	TOPIC:6 CAUCHY'S INTEGRAL TEST, THE P-SERIES
	Cauchy's Integral Test **Statement:** Let $f(1)+f(2)+f(3)+f(4)+\cdots+f(n)+\cdots$be a positive term series. And let f be a continuous, decreasing function defined on $[1,\infty)$. Then the series $\sum_{n=1}^{\infty} f(n)$ converges if $\int_1^{\infty} f(x)dx$ is finite or diverges if $\int_1^{\infty} f(x)dx$ is infinite **The p-series:** **The p-series $\sum_{n=1}^{\infty}\frac{1}{n^p}$ (p - a real constant)** **Converges if p > 1and diverges if p**≤ 1

	NUMERICAL QUESTIONS
1	Test the convergence of $\sum_{n=1}^{\infty}\frac{1}{n^{3/2}}$ **Solution:** $\sum_{n=1}^{\infty} u_n = \sum_{n=1}^{\infty}\frac{1}{n^{3/2}}$ Using p-series test, The series $\sum_{n=1}^{\infty}\frac{1}{n^{3/2}}$ is Convergent as $p = \frac{3}{2} > 1$ **Answer**
2	Check whether the series $\sum\frac{1}{\sqrt{n}}$ is Convergent or Divergent **Solution:** $\sum_{n=1}^{\infty} u_n = \sum_{n=1}^{\infty}\frac{1}{\sqrt{n}}$ Using p-series test, The series $\sum_{n=1}^{\infty}\frac{1}{\sqrt{n}}$ is Divergent as $p = \frac{1}{2} < 1$ **Answer**
3	State the Integral Test and determine the convergence of $\sum_{n=1}^{\infty}\frac{2\tan^{-1} n}{1+n^2}$ **Solution:** **Cauchy's Integral Test** **Statement:** Let $f(1) + f(2) + f(3) + f(4) + \cdots + f(n) + \cdots$be a positive term series. And let f be a continuous, decreasing function defined on $[1, \infty)$. Then the series $\sum_{n=1}^{\infty} f(n)$ converges if $\int_1^{\infty} f(x)dx$ is finite or diverges if $\int_1^{\infty} f(x)dx$ is infinite **To determine Convergence of the series:** $u_n = \frac{2\tan^{-1} n}{1+n^2} = f(n)$ $f(x) = \frac{2\tan^{-1} x}{1+x^2}, x \geq 1$ f is continuous, positive and decreasing on $[1,\infty)$

Exploring Emerging Technologies

$$\int_1^\infty f(x)dx = \int_1^\infty \frac{2\tan^{-1}x}{1+x^2}dx$$

$$= \lim_{u\to\infty}\int_1^u \frac{2\tan^{-1}x}{1+x^2}dx$$

Putting $\tan^{-1}x = t$, $\frac{dx}{1+x^2} = dt$

When $x = 1$, $t = \tan^{-1}(1) = \frac{\pi}{4}$

When $x = u$, $t = \tan^{-1}(u)$

$$\int_{x=1}^\infty f(x)dx = \lim_{u\to\infty}\int_{t=\frac{\pi}{4}}^{\tan^{-1}(u)} 2tdt$$

$$= \lim_{u\to\infty}\left[\frac{2t^2}{2}\right]_{t=\frac{\pi}{4}}^{\tan^{-1}(u)}$$

$$= \lim_{u\to\infty}\left[(\tan^{-1}u)^2 - \frac{\pi^2}{16}\right]$$

$$= (\tan^{-1}\infty)^2 - \frac{\pi^2}{16}$$

$$= \frac{\pi^2}{4} - \frac{\pi^2}{16}$$

$$= \boldsymbol{\frac{3\pi^2}{16}}, \boldsymbol{finite}$$

Hence, by Cauchy's Integral Test, the series is convergent. **Answer**

4 Test the convergence of the series $\sum_{n=2}^\infty \frac{1}{n\log n}$

Solution:

$$u_n = \frac{1}{n\log n} = f(n)$$

$$f(x) = \frac{1}{x\log x}, x \geq 2$$

f is continuous, positive and decreasing on $[2,\infty)$

$$\int_2^\infty f(x)dx = \int_2^\infty \frac{1}{x\log x}dx$$

	$= \lim_{u\to\infty} \int_2^u \frac{1}{x\log x} dx$ $= \lim_{u\to\infty} [\log(\log x)]_{x=2}^{x=u} \quad \left[\because \int \frac{f'(x)}{f(x)} dx = \log\lvert f(x)\rvert, here\ f(x) = \log x\ and\ f'(x) = \frac{1}{x}\right]$ $= \lim_{u\to\infty} (\log(\log u) - \log(\log 2)) \to \infty, \boldsymbol{infinite}$ **Hence, by Cauchy's Integral Test, the series is divergent.** **Answer**
5	Write the statement of Cauchy's integral test. Test the convergence of the following series $\sum_{n=2}^{\infty} \frac{1}{n(logn)^a}$ $, for\ 0 \le a \le 1.$ **Solution:** $u_n = \frac{1}{n(\log n)^a} = f(n)$ $f(x) = \frac{1}{x(\log x)^a}, \quad x \ge 2$ f is continuous, positive and decreasing on [2,∞) $\int_2^\infty f(x)dx = \int_2^\infty \frac{1}{x(\log x)^a} dx$ $= \lim_{u\to\infty} \int_2^u \frac{1}{x(\log x)^a} dx$ Take $\log x = t, \ \therefore \frac{1}{x} dx = dt$ $When\ x = 2, \qquad t = \log 2$ $When\ x = u, \qquad t = \log u$ $\int_{x=2}^\infty f(x)dx = \lim_{u\to\infty} \int_{t=\log 2}^{\log u} \frac{1}{t^a} dt$ $= \lim_{u\to\infty} \int_{t=\log 2}^{\log u} t^{-a} dt$ $= \lim_{u\to\infty} \left[\frac{t^{-a+1}}{-a+1}\right]_{t=\log 2}^{\log u}$

	$= \lim_{u\to\infty} \frac{1}{1-a}[(\log u)^{1-a} - (\log 2)^{1-a}] \to \infty, infinite \quad as\ 0 \le a \le 1 \Rightarrow 1-a > 0$ **Hence, by Cauchy's Integral Test, the series is divergent.** **Answer**
6	Test the convergence of the series $\sum ne^{-n^2}$ **Solution:** $u_n = ne^{-n^2} = f(n)$ $f(x) = xe^{-x^2}, \ x \ge 1$ f is continuous, positive and decreasing on $[1,\infty)$ $\int_{x=1}^{\infty} f(x)dx = \int_{x=1}^{\infty} xe^{-x^2}dx$ $= \lim_{u\to\infty} \int_{x=1}^{u} xe^{-x^2}dx$ Take $x^2 = t, \quad \therefore 2xdx = dt \Rightarrow xdx = \frac{dt}{2}$ $When\ x = 1, \quad t = 1$ $When\ x = u, \quad t = u^2$ $\int_{x=1}^{\infty} f(x)dx = \lim_{u\to\infty} \int_{t=1}^{u^2} e^{-t}\left(\frac{dt}{2}\right)$ $= \lim_{u\to\infty} \left[\frac{e^{-t}}{-2}\right]_{t=1}^{u^2}$ $= \lim_{u\to\infty} \frac{1}{(-2)}\left[e^{(-u^2)} - e^{(-1)}\right]$ $= \lim_{u\to\infty} \frac{1}{(-2)}\left[\frac{1}{e^{u^2}} - \frac{1}{e}\right]$ $= \left(-\frac{1}{2}\right)\left[0 - \frac{1}{e}\right]$ $= \frac{1}{2e}, finite$ **Hence, by Cauchy's Integral Test, the series is convergent.** **Answer**

7	Test the convergence and Divergence of the following series $\sum_{n=1}^{\infty} \frac{1}{n(1+log^2 n)}$ **Solution:** $u_n = \frac{1}{n(1+log^2 n)} = f(n)$ $f(x) = \frac{1}{x(\log x)^a}$, $x \geq 1$ f is continuous, positive and decreasing in $[1,\infty)$ $\int_1^{\infty} f(x)dx = \int_1^{\infty} \frac{1}{x(1+log^2 x)} dx$ $= \lim_{u\to\infty} \int_{x=1}^{u} \frac{1}{x(1+log^2 x)} dx$ Take $\log x = t$, $\therefore \frac{1}{x}\, dx = dt$ $When\ x = 1,\quad t = \log 1 = 0$ $When\ x = u,\quad t = \log u$ $\int_{x=1}^{\infty} f(x)dx = \lim_{u\to\infty} \int_{t=0}^{\log u} \frac{1}{1+t^2} dt$ $= \lim_{u\to\infty} \int_{t=0}^{\log u} [\tan^{-1} t]_{t=0}^{\log u}$ $= \lim_{u\to\infty} [\tan^{-1}(\log u) - \tan^{-1}(0)]$ $= \frac{\pi}{2}, finite$ **Hence, by Cauchy's Integral Test, the series is convergent.** **Answer**
8	Does $\sum_{n=1}^{\infty} e^{-n}$ series converge or diverge? **Solution:** $u_n = e^{-n} = f(n)$ $f(x) = e^{-x}$, $x \geq 1$ f is continuous, positive and decreasing on $[1,\infty)$

$$\int_{x=1}^{\infty} f(x)dx = \int_{x=1}^{\infty} e^{-x}dx$$

$$= \lim_{u\to\infty} \int_{x=1}^{u} e^{-x}dx$$

$$= \lim_{u\to\infty} \left[\frac{e^{-x}}{-1}\right]_{x=1}^{u}$$

$$= \lim_{u\to\infty} \frac{1}{(-1)}\left[e^{(-u)} - e^{(-1)}\right]$$

$$= \lim_{u\to\infty} \frac{1}{(-1)}\left[\frac{1}{e^{u}} - \frac{1}{e}\right]$$

$$= (-1)\left[0 - \frac{1}{e}\right]$$

$$= \frac{1}{e}, finite$$

Hence, by Cauchy's Integral Test, the series is convergent. **Answer**

9 Test the convergence of $\sum_{n=3}^{\infty} \frac{1}{n\log n\sqrt{log^2 n-1}}$

Solution:

$$u_n = \frac{1}{n\log n\sqrt{log^2 n - 1}} = f(n)$$

$$f(x) = \frac{1}{x\log x\sqrt{log^2 x - 1}}, \quad x \geq 3$$

f is continuous, positive and decreasing on $[3,\infty)$

$$\int_{3}^{\infty} f(x)dx = \int_{3}^{\infty} \frac{1}{x\log x\sqrt{log^2 x - 1}}dx$$

$$= \lim_{u\to\infty} \int_{3}^{u} \frac{1}{x\log x\sqrt{log^2 x - 1}}dx$$

Take $\log x = t, \ \therefore \frac{1}{x}\, dx = dt$

$When\ x = 3, \qquad t = \log 3$

$When\ x = u, \qquad t = \log u$

$$\int_{x=3}^{\infty} f(x)dx = \lim_{u\to\infty} \int_{t=\log 3}^{\log u} \frac{1}{t\sqrt{t^2-1}}dt$$

$$= \lim_{u\to\infty} [\sec^{-1} t]_{t=\log 3}^{\log u}$$

$$= \lim_{u\to\infty} [\sec^{-1}(\log u) - \sec^{-1}(\log 3)]$$

$$= \sec^{-1}(\log \infty) - \sec^{-1}(\log 3)$$

$$= \frac{\pi}{2} - \sec^{-1}(\log 3), finite$$

Hence, by Cauchy's Integral Test, the series is convergent. **Answer**

TOPIC:7 THE COMPARISON TEST, THE LIMIT COMPARISON TEST

Statement:

Let $\sum u_n$ and $\sum v_n$ be two positive term series such that

$$\lim_{n\to\infty} \frac{u_n}{v_n} = 1 \text{ (finite and non-zero)}$$

then both series converge or diverge together.

Note: The test series $\sum v_n$ is known as auxiliary series and is obtained by

$$v_n = \frac{highest\ power\ term\ in\ numerator\ of\ u_n}{highest\ power\ term\ in\ denominator\ of\ u_n}$$

SHORT QUESTIONS

1 For which values of p do the series $\sum_{n=1}^{\infty} \frac{n+1}{n^p}$ is convergent?

Solution:

Here,

$$u_n = \frac{n+1}{n^p}$$

Take $v_n = \frac{n}{n^p} = \frac{1}{n^{p-1}}$

$$\lim_{n\to\infty} \frac{u_n}{v_n} = \lim_{n\to\infty} \left(\frac{n+1}{n^p}\right) \times n^{p-1} = \lim_{n\to\infty} \left(\frac{n+1}{n}\right) = \left(1+\frac{1}{n}\right) = 1+0$$

	$= 1, finite\ \ and\ non-zero$ Hence, by Limit Comparison test, both $\sum u_n\ and\ \sum v_n$ converge or diverge together. But $\sum v_n = \sum \frac{1}{n^{p-1}}$ **is convergent when** $p-1>1\ \ i.e. p>2$ $\therefore \sum u_n$ **is convergent when** $p>2$. **Answer**
2	The series $\sum_{n=1}^{\infty} \frac{\sin n}{n^2}$ **Solution:** $u_n = \frac{\sin n}{n^2}$ $\sin n \le 1$ $\Rightarrow \frac{\sin n}{n^2} \le \frac{1}{n^2}$ Let $v_n = \frac{1}{n^2}$ then $u_n \le v_n\ for\ all\ n \ge 1$ But $\sum v_n = \sum \frac{1}{n^2}$ is a $p-series$ with $p=2>1$, so it is convergent. $\therefore$ **By Direct Comparison Test,** $\sum u_n$ **is also convergent.** **Answer**
	NUMERICAL QUESTIONS
	TO FIND VALUE OF P FOR WHICH SERIES CONVERGES
1	For which values of p do the series $\sum_{n=1}^{\infty} \frac{n^p}{\sqrt{n+1}+\sqrt{n}}$ is convergent. **Solution:** Here, $u_n = \frac{n^p}{\sqrt{n+1}+\sqrt{n}}$ Take $v_n = \frac{n^P}{\sqrt{n}} = \frac{1}{(n)^{\frac{1}{2}-p}}$ $\lim_{n\to\infty} \frac{u_n}{v_n} = \lim_{n\to\infty} \frac{n^p}{\sqrt{n+1}+\sqrt{n}} \times (n)^{\frac{1}{2}-p} = \lim_{n\to\infty} \left(\frac{\sqrt{n}}{\sqrt{n+1}+\sqrt{n}}\right) = \left(\frac{1}{\sqrt{1+0}+1}\right)$ $= \frac{1}{2}, finite\ \ and\ non-zero$ Hence, by Limit Comparison test, both $\sum u_n\ and\ \sum v_n$ converge or diverge together.

	But $\sum v_n = \sum \frac{1}{(n)^{\frac{1}{2}-p}}$ **is convergent when** $\frac{1}{2} - p > 1 \;\; i.e. p < \left(-\frac{1}{2}\right)$ $\therefore \sum u_n$ **is convergent when** $p < \left(-\frac{1}{2}\right)$ **Answer**
2	For which values of p do the series $\frac{2}{1^p} + \frac{3}{2^p} + \frac{4}{3^p} + \ldots$ is convergent. **Solution:** Here, $u_n = \frac{n+1}{n^p}$ Take $v_n = \frac{n}{n^p} = \frac{1}{n^{P-1}}$ $\lim_{n\to\infty} \frac{u_n}{v_n} = \lim_{n\to\infty} \left(\frac{n+1}{n^p}\right) \times n^{P-1} = \lim_{n\to\infty} \left(\frac{n+1}{n}\right) = \left(1 + \frac{1}{n}\right) = 1 + 0$ $= 1, finite \;\; and \; non - zero$ Hence, by Limit Comparison test, both $\sum u_n \; and \; \sum v_n$ converge or diverge together. But $\sum v_n = \sum \frac{1}{n^{p-1}}$ **is convergent when** $p - 1 > 1 \;\; i.e. p > 2$ $\therefore \sum u_n$ **is convergent when** $p > 2$. **Answer**
	USE OF CONJUGATE SURDS
3	Discuss the convergence of $\sum_{n=1}^{\infty} [\sqrt{n^2+1} - n]$ **Solution:** $u_n = \sqrt{n^2+1} - n$ $= \sqrt{n^2+1} - n \times \frac{\sqrt{n^2+1}+n}{\sqrt{n^2+1}+n} (Conjugate \; surd)$ $= \frac{(n^2+1) - n^2}{\sqrt{n^2+1}+n}$ $= \frac{1}{\sqrt{n^2+1}+n}$ Take $v_n = \frac{1}{n}$

	$\lim_{n\to\infty}\frac{u_n}{v_n}=\lim_{n\to\infty}\left(\frac{1}{\sqrt{n^2+1}+n}\right)\times n=\lim_{n\to\infty}\left(\frac{1}{\sqrt{1+\frac{1}{n^2}}+1}\right)=\left(\frac{1}{\sqrt{1+0}+1}\right)$ $=\frac{1}{2}, finite\ \ and\ non-zero$ Hence, by Limit Comparison test, both $\sum u_n$ *and* $\sum v_n$ converge or diverge together. But $\sum v_n=\sum\frac{1}{n}$ **is a $p-series$ with $p=1$, so it is divergent.** $\therefore\ \sum u_n$ **is also divergent.** **Answer**
4	Test the convergence of the series $\sum_{n=1}^{\infty}[\sqrt{n^4+1}-\sqrt{n^4-1}]$ **Solution:** $u_n=\sqrt{n^4+1}-\sqrt{n^4-1}$ $=\sqrt{n^4+1}-\sqrt{n^4-1}\times\frac{\sqrt{n^4+1}+\sqrt{n^4-1}}{\sqrt{n^4+1}+\sqrt{n^4-1}}(Conjugate\ surd)$ $=\frac{(n^4+1)-(n^4-1)}{\sqrt{n^4+1}+\sqrt{n^4-1}}$ $=\frac{2}{\sqrt{n^4+1}+\sqrt{n^4-1}}$ Take $v_n=\frac{1}{n^2}$ $\lim_{n\to\infty}\frac{u_n}{v_n}=\lim_{n\to\infty}\frac{2}{\sqrt{n^4+1}+\sqrt{n^4-1}}\times n^2=\lim_{n\to\infty}\frac{2}{\sqrt{1+\frac{1}{n^4}}+\sqrt{1-\frac{1}{n^4}}}=\frac{2}{1+1}$ $=1, finite\ and\ non-zero$ Hence, by Limit Comparison test, both $\sum u_n$ *and* $\sum v_n$ converge or diverge together. But $\sum v_n=\sum\frac{1}{n^2}$ **is a $p-series$ with $p=2>1$, so it is convergent.** $\therefore\sum u_n$ **is also convergent.** **Answer**
5	Test the convergence of the series $\sum_{n=1}^{\infty}\sqrt{n+1}-\sqrt{n}$ **Solution:** $u_n=\sqrt{n+1}-\sqrt{n}$

$$= \sqrt{n+1} - \sqrt{n} \times \frac{\sqrt{n+1}+\sqrt{n}}{\sqrt{n+1}+\sqrt{n}} (Conjugate\ surd)$$

$$= \frac{(n+1)-n}{\sqrt{n+1}+\sqrt{n}}$$

$$= \frac{1}{\sqrt{n+1}+\sqrt{n}}$$

Take $v_n = \frac{1}{\sqrt{n}}$

$$\lim_{n\to\infty} \frac{u_n}{v_n} = \lim_{n\to\infty} \left(\frac{1}{\sqrt{n+1}+\sqrt{n}}\right) \times \sqrt{n} = \lim_{n\to\infty} \left(\frac{1}{\sqrt{1+\frac{1}{n}}+1}\right) = \left(\frac{1}{\sqrt{1+0}+1}\right)$$

$$= \frac{1}{2}, finite\ \ and\ non-zero$$

Hence, by Limit Comparison test, both $\sum u_n\ and\ \sum v_n$ converge or diverge together.

But $\sum v_n = \sum \frac{1}{\sqrt{n}} = \sum \frac{1}{n^{1/2}}$ **is a $p-series$ with $p = \frac{1}{2} < 1$, so it is divergent.**

$\therefore$ $\sum u_n$ **is also divergent.** **Answer**

USE OF BINOMIAL EXPANSION

6 Test for the convergence the series; $\sum_{n=1}^{\infty} \sqrt[3]{n^3+1} - n$

Solution:

$$u_n = \sqrt[3]{n^3+1} - n$$

$$= (n^3+1)^{\frac{1}{3}} - n$$

$$= n\left(1+\frac{1}{n^3}\right)^{\frac{1}{3}} - n$$

$$= n\left[1 + \frac{1}{3}\frac{1}{n^3} + \frac{\frac{1}{3}\left(\frac{1}{3}-1\right)}{2!}\left(\frac{1}{n^3}\right)^2 + \frac{\frac{1}{3}\left(\frac{1}{3}-1\right)\left(\frac{1}{3}-2\right)}{3!}\left(\frac{1}{n^3}\right)^3 + \cdots\right] - n$$

$$\left(\because (1+x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + \frac{n(n-1)(n-2)}{3!}x^3 + \cdots\right)$$

$$= \frac{1}{3}\frac{1}{n^2} - \frac{1}{3^2}\frac{1}{n^5} + \frac{5}{3^4}\frac{1}{n^8} - \cdots$$

	$= \frac{1}{n^2}\left(\frac{1}{3} - \frac{1}{3^2}\frac{1}{n^3} + \frac{5}{3^4}\frac{1}{n^6} - \cdots\right)$ Take $v_n = \frac{1}{n^2}$ $\lim_{n\to\infty}\frac{u_n}{v_n} = \lim_{n\to\infty}\left(\frac{1}{n^2}\left(\frac{1}{3} - \frac{1}{3^2}\frac{1}{n^3} + \frac{5}{3^4}\frac{1}{n^6} - \cdots\right)\right) \times n^2 = \frac{1}{3}, finite\ \ and\ non-zero$ Hence, by Limit Comparison test, both $\sum u_n$ *and* $\sum v_n$ converge or diverge together. But $\sum v_n = \sum \frac{1}{n^2}$ **is a $p-series$ with $p = 2 > 1$, so it is convergent.** $\therefore$ $\sum u_n$ **is also convergent.** **Answer**
	LOGARITHMIC FUNCTION
7	Test the convergence of $\sum_{n=1}^{\infty}\frac{\ln n}{n^{3/2}}$ **Solution:** Because $ln\, n$ (Natural logarithm) grows more slowly than n^c, for any positive constant c, we have $\ln n < n^c$, c is a positive constant $\Rightarrow \ln n < n^{1/4}$ $(As\ taking\ c = 1/4)$ $\frac{\ln n}{n^{3/2}} < \frac{n^{1/4}}{n^{3/2}} = \frac{1}{n^{(3/2)-(1/4)}} = \frac{1}{n^{5/4}}\ for\ n\ sufficiently\ large$ Taking $u_n = \frac{\ln n}{n^{3/2}}$ and $v_n = \frac{1}{n^{5/4}}$ $\lim_{n\to\infty}\frac{u_n}{v_n} = \lim_{n\to\infty}\frac{\ln n}{n^{\frac{3}{2}}} \times n^{\frac{5}{4}}$ $= \lim_{n\to\infty}\frac{\ln n}{n^{\frac{1}{4}}}$ $= \lim_{n\to\infty}\frac{\frac{1}{n}}{\left(\frac{1}{4}\right)n^{(-3/4)}}\ \left[L'Hospital\ Rule\ for\ \frac{\infty}{\infty}\ form\right]$ $= \lim_{n\to\infty}\frac{4}{n^{1/4}}$ $= 0$

	But $\sum v_n = \sum \frac{1}{n^{5/4}}$ **is a** $p-series$ **with** $p = \frac{5}{4} > 1$, **so it is convergent.** $\therefore$ By Limit Comparison test, $\sum u_n$ **is also convergent.** **Answer**
8	Check the convergence of the series $\sum_{n=1}^{\infty} \frac{(\ln n)^3}{n^3}$ **Solution:** Because $ln\, n$ (Natural logarithm) grows more slowly than n^c, for any positive constant c, we have $\ln n < n^c$, c is a positive constant $\Rightarrow \ln n < n^{1/4}$ $(As\ taking\ c = 1/4)$ $(\ln n)^3 < \left(n^{1/4}\right)^3 = n^{3/4}$ $\frac{(\ln n)^3}{n^3} < \frac{n^{3/4}}{n^3} = \frac{1}{n^{3-(3/4)}} = \frac{1}{n^{9/4}}$ Taking $u_n = \frac{(\ln n)^3}{n^3}$ and $v_n = \frac{1}{n^{9/4}}$ $\lim_{n\to\infty} \frac{u_n}{v_n} = \lim_{n\to\infty} \frac{(\ln n)^3}{n^3} \times n^{9/4}$ $= \lim_{n\to\infty} \frac{(\ln n)^3}{n^{3-(9/4)}}$ $= \lim_{n\to\infty} \frac{(\ln n)^3}{n^{3/4}}$ $= \lim_{n\to\infty} \frac{3\ln n \left(\frac{1}{n}\right)}{\left(\frac{3}{4}\right) n^{(-1/4)}}$ $\left[L'Hospital\ Rule\ for\ \frac{\infty}{\infty}\ form\right]$ $= \lim_{n\to\infty} \frac{4\ln n}{n^{1-(1/4)}}$ $= \lim_{n\to\infty} \frac{4\ln n}{n^{\frac{3}{4}}}$ $= \lim_{n\to\infty} \frac{4\frac{1}{n}}{\left(\frac{3}{4}\right) n^{(-1/4)}}$ $\left[L'Hospital\ Rule\ for\ \frac{\infty}{\infty}\ form\right]$ $= \lim_{n\to\infty} \left(\frac{16}{3}\right) \frac{1}{n^{3/4}}$

	$= 0$ But $\sum v_n = \sum \frac{1}{n^{9/4}}$ **is a** $p - series$ **with** $p = \frac{9}{4} > 1$, **so it is convergent.** $\therefore$ By Limit Comparison test, $\sum u_n$ **is also convergent.** **Answer**
	TRIGONOMETRIC FUNCTION
9	Using comparison test, discuss the convergence of $\sum_{n=1}^{\infty} \frac{1}{n} \sin\left(\frac{1}{n}\right)$ **Solution:** $u_n = \frac{1}{n} \sin\left(\frac{1}{n}\right)$ $= \frac{1}{n}\left[\left(\frac{1}{n}\right) - \frac{\left(\frac{1}{n}\right)^3}{3!} + \frac{\left(\frac{1}{n}\right)^5}{5!} - \frac{\left(\frac{1}{n}\right)^7}{7!} \ldots\right]\left(As \text{ Expansion of } \sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots\right)$ $= \frac{1}{n^2}\left[1 - \frac{\left(\frac{1}{n}\right)^2}{3!} + \frac{\left(\frac{1}{n}\right)^4}{5!} - \frac{\left(\frac{1}{n}\right)^6}{7!} \ldots\right]$ Take $v_n = \frac{1}{n^2}$ $\lim_{n\to\infty} \frac{u_n}{v_n} = \lim_{n\to\infty}\left(\frac{1}{n}\sin\left(\frac{1}{n}\right)\right) \times n^2 = \lim_{n\to\infty}\left(\frac{\sin\left(\frac{1}{n}\right)}{\left(\frac{1}{n}\right)}\right) = 1, finite \text{ and } non - zero$ Hence, by Limit Comparison test, both $\sum u_n$ *and* $\sum v_n$ converge or diverge together. But $\sum v_n = \sum \frac{1}{n^2}$ **is a** $p - series$ **with** $p = 2 > 1$, **so it is convergent.** $\therefore$ $\sum u_n$ **is also convergent.** **Answer**
10	Examine the convergence of $\sum_{n=1}^{\infty} \sin\left(\frac{1}{n}\right)$ **Solution:** $u_n = \sin\left(\frac{1}{n}\right)$ $= \left(\frac{1}{n}\right) - \frac{\left(\frac{1}{n}\right)^3}{3!} + \frac{\left(\frac{1}{n}\right)^5}{5!} - \frac{\left(\frac{1}{n}\right)^7}{7!} \ldots\left(As \text{ Expansion of } \sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots\right)$

	$= \frac{1}{n}\left[1 - \frac{\left(\frac{1}{n}\right)^2}{3!} + \frac{\left(\frac{1}{n}\right)^4}{5!} - \frac{\left(\frac{1}{n}\right)^6}{7!} \dots\right]$ Take $v_n = \frac{1}{n}$ $\lim_{n\to\infty} \frac{u_n}{v_n} = \lim_{n\to\infty}\left(\sin\left(\frac{1}{n}\right)\right) \times n = \lim_{n\to\infty}\left(\frac{\sin\left(\frac{1}{n}\right)}{\left(\frac{1}{n}\right)}\right) = 1, finite\ \ and\ non-zero$ Hence, by Limit Comparison test, both $\sum u_n$ *and* $\sum v_n$ converge or diverge together. But $\sum v_n = \sum \frac{1}{n}$ **is a $p-series$ with $p = 1$, so it is divergent.** $\therefore$ $\sum u_n$ **is also divergent.** **Answer**
	USE OF DIRECT COMPARISION TEST
11	Discuss the convergence of the following series $\sum \frac{2+3\cos n}{n^3}$ **Solution:** $u_n = \frac{2+3\cos n}{n^3}$ $\cos n \le 1$ $\Rightarrow 3\cos n \le 3$ $\Rightarrow 2 + 3\cos n \le 2 + 3$ $\Rightarrow \frac{2+3\cos n}{n^3} \le \frac{5}{n^3}\ for\ n \ge 1$ Take $v_n = \frac{5}{n^3}$ then $u_n \le v_n$ But $\sum v_n = \sum \frac{5}{n^3} = 5\sum \frac{1}{n^3}$ **is a $p-series$ with $p = 3 > 1$, so it is convergent.** $\therefore$ ***By Direct Comparison Test,*** $\sum u_n$ **is also convergent.** **Answer**
	USE OF GEOMETRIC SERIES
12	Determine the convergence of $\sum_{n=1}^{\infty} \frac{1}{5^n - 1}$ **Solution:** $u_n = \frac{1}{5^n - 1}$

	Take $v_n = \frac{1}{5^n}$ $\lim_{n\to\infty} \frac{u_n}{v_n} = \lim_{n\to\infty} \left(\frac{1}{5^n - 1}\right) \times 5^n = \lim_{n\to\infty} \left(\frac{1}{1 - \frac{1}{5^n}}\right) = 1, finite \ \ and\ non - zero$ Hence, by Limit Comparison test, both $\sum u_n$ *and* $\sum v_n$ converge or diverge together. But $\sum v_n = \sum \frac{1}{5^n}$ **is a *geometric series with* $r = \frac{1}{5} < 1$, so it is convergent.** $\therefore$ $\sum u_n$ **is also convergent.** **Answer**
	ALGEBRAIC FUNCTION
13	Discuss the convergence of $\sum_{n=1}^{\infty} \frac{2n+1}{n^2+2n+1}$ Does $\sum_{n=1}^{\infty} \frac{2n+1}{n^2+2n+1}$ converge or diverge? **Solution:** $u_n = \frac{2n+1}{(n+1)^2} = \frac{2n+1}{n^2+2n+1}$ Take $v_n = \frac{n}{n^2} = \frac{1}{n}$ $\lim_{n\to\infty} \frac{u_n}{v_n} = \lim_{n\to\infty} \frac{2n+1}{n^2+2n+1} \times n = \lim_{n\to\infty} \frac{2+\frac{1}{n}}{1+\frac{2}{n}+\frac{1}{n^2}} = 2, finite\ and\ non - zero$ Hence, by Limit Comparison test, both $\sum u_n$ *and* $\sum v_n$ converge or diverge together. But $\sum v_n = \sum \frac{1}{n}$ **is a *p − series* with $p = 1$, so it is divergent.** $\therefore \sum u_n$ **is also divergent.** **Answer**
14	Test the convergence of the series $\sum_{n=1}^{\infty} \frac{n+1}{n^3+3}$ **Solution:** $u_n = \frac{n+1}{n^3+3}$ Take $v_n = \frac{n}{n^3} = \frac{1}{n^2}$ $\lim_{n\to\infty} \frac{u_n}{v_n} = \lim_{n\to\infty} \frac{n+1}{n^3+3} \times n^2 = \lim_{n\to\infty} \frac{1+\frac{1}{n}}{1+\frac{3}{n}} = 1, finite\ and\ non - zero$ Hence, by Limit Comparison test, both $\sum u_n$ *and* $\sum v_n$ converge or diverge together.

	But $\sum v_n = \sum \frac{1}{n^2}$ **is a $p-series$ with $p = 2 > 1$, so it is convergent.** $\therefore \sum u_n$ **is also convergent.** **Answer**
15	Using comparison test, discuss the convergence of $\sum_{n=1}^{\infty} \frac{\sqrt{n}-1}{n^2+1}$ **Solution:** $u_n = \frac{\sqrt{n}-1}{n^2+1}$ Take $v_n = \frac{\sqrt{n}}{n^2} = \frac{1}{n^{2-(\frac{1}{2})}} = \frac{1}{n^{3/2}}$ $\lim_{n\to\infty} \frac{u_n}{v_n} = \lim_{n\to\infty} \frac{\sqrt{n}-1}{n^2+1} \times n^{3/2} = \lim_{n\to\infty} \frac{1-\frac{1}{\sqrt{n}}}{1+\frac{1}{n^2}} = 1, finite\ and\ non-zero$ Hence, by Limit Comparison test, both $\sum u_n$ *and* $\sum v_n$ converge or diverge together. But $\sum v_n = \sum \frac{1}{n^{3/2}}$ **is a $p-series$ with $p = \frac{3}{2} > 1$, so it is convergent.** $\therefore \sum u_n$ **is also convergent.** **Answer**
16	Determine whether series converges or diverges $\sum_{n=1}^{\infty} \frac{1}{n^2+n+1}$. **Solution:** $u_n = \frac{1}{n^2+n+1}$ Take $v_n = \frac{1}{n^2}$ $\lim_{n\to\infty} \frac{u_n}{v_n} = \lim_{n\to\infty} \frac{1}{n^2+n+1} \times n^2 = \lim_{n\to\infty} \frac{1}{1+\frac{1}{n}+\frac{1}{n^2}} = 1, finite\ and\ non-zero$ Hence, by Limit Comparison test, both $\sum u_n$ *and* $\sum v_n$ converge or diverge together. But $\sum v_n = \sum \frac{1}{n^2}$ **is a $p-series$ with $p = 2 > 1$, so it is convergent.** $\therefore \sum u_n$ **is also convergent.** **Answer**
17	State the p-series test. Test the convergence of series $\sum_{n=0}^{\infty} \frac{n+1}{n^3-3n+2}$ **Solution:** $u_n = \frac{n+1}{n^3-3n+2}$

	Take $v_n = \frac{n}{n^3} = \frac{1}{n^2}$ $\lim_{n\to\infty} \frac{u_n}{v_n} = \lim_{n\to\infty} \frac{n+1}{n^3-3n+2} \times n^2 = \lim_{n\to\infty} \frac{1+\frac{1}{n}}{1-\frac{3}{n^2}+\frac{2}{n^3}} = 1, finite\ and\ non-zero$ Hence, by Limit Comparison test, both $\sum u_n$ *and* $\sum v_n$ converge or diverge together. But $\sum v_n = \sum \frac{1}{n^2}$ **is a $p-series$ with $p = 2 > 1$, so it is convergent.** $\therefore \sum u_n$ **is also convergent.** **Answer**
18	Test the convergence of the series $\sum_{n=1}^{\infty} \frac{1}{n^2+1}$ **Solution:** $u_n = \frac{1}{n^2+1}$ Take $v_n = \frac{1}{n^2}$ $\lim_{n\to\infty} \frac{u_n}{v_n} = \lim_{n\to\infty} \frac{1}{n^2+1} \times n^2 = \lim_{n\to\infty} \frac{1}{1+\frac{1}{n^2}} = 1, finite\ and\ non-zero$ Hence, by Limit Comparison test, both $\sum u_n$ *and* $\sum v_n$ converge or diverge together. But $\sum v_n = \sum \frac{1}{n^2}$ **is a $p-series$ with $p = 2 > 1$, so it is convergent.** $\therefore \sum u_n$ **is also convergent.** **Answer**
19	Check for convergence/divergence $\sum_{n=1}^{\infty} \frac{5n^3-3n}{n^2(n-2)(n^2+5)}$ **Solution:** $u_n = \frac{5n^3-3n}{n^2(n-2)(n^2+5)}$ Take $v_n = \frac{n^3}{n^2 n n^2} = \frac{1}{n^2}$ $\lim_{n\to\infty} \frac{u_n}{v_n} = \lim_{n\to\infty} \frac{5n^3-3n}{n^2(n-2)(n^2+5)} \times n^2 = \lim_{n\to\infty} \frac{5-\frac{3}{n^2}}{\left(1-\frac{2}{n}\right)(1+\frac{5}{n^2})}$ $= 5, finite\ and\ non-zero$ Hence, by Limit Comparison test, both $\sum u_n$ *and* $\sum v_n$ converge or diverge together. But $\sum v_n = \sum \frac{1}{n^2}$ **is a $p-series$ with $p = 2 > 1$, so it is convergent.**

	$\therefore \sum u_n$ **is also convergent.** **Answer**
20	Determine whether the series $\sum_{n=1}^{\infty} \frac{(n-1)}{n^3+3}$ converges or diverges. **Solution:** $u_n = \frac{(n-1)}{n^3+3}$ Take $v_n = \frac{n}{n^3} = \frac{1}{n^2}$ $\lim_{n\to\infty} \frac{u_n}{v_n} = \lim_{n\to\infty} \frac{(n-1)}{n^3+3} \times n^2 = \lim_{n\to\infty} \frac{1-\frac{1}{n}}{1+\frac{3}{n^3}} = 1, finite\ and\ non-zero$ Hence, by Limit Comparison test, both $\sum u_n$ *and* $\sum v_n$ converge or diverge together. But $\sum v_n = \sum \frac{1}{n^2}$ **is a** $p-series$ **with** $p = 2 > 1$**, so it is convergent.** $\therefore \sum u_n$ **is also convergent.** **Answer**
21	Test the convergence of $\sum_{n=1}^{\infty} \frac{3n^2+5n}{7+n^4}$ **Solution:** $u_n = \frac{3n^2+5n}{7+n^4}$ Take $v_n = \frac{n^2}{n^4} = \frac{1}{n^2}$ $\lim_{n\to\infty} \frac{u_n}{v_n} = \lim_{n\to\infty} \frac{3n^2+5n}{7+n^4} \times n^2 = \lim_{n\to\infty} \frac{3+\frac{5}{n}}{\frac{7}{n^4}+1} = 3, finite\ and\ non-zero$ Hence, by Limit Comparison test, both $\sum u_n$ *and* $\sum v_n$ converge or diverge together. But $\sum v_n = \sum \frac{1}{n^2}$ **is a** $p-series$ **with** $p = 2 > 1$**, so it is convergent.** $\therefore \sum u_n$ **is also convergent.** **Answer**
22	Examine the convergence of $\sum_{n=1}^{\infty} \frac{2n^2+3n}{5+n^5}$ **Solution:** $u_n = \frac{2n^2+3n}{5+n^5}$ Take $v_n = \frac{n^2}{n^5} = \frac{1}{n^3}$

	$\lim_{n\to\infty}\frac{u_n}{v_n}=\lim_{n\to\infty}\frac{2n^2+3n}{5+n^5}\times n^3=\lim_{n\to\infty}\frac{2+\frac{3}{n}}{\frac{5}{n^5}+1}=2, finite\ and\ non-zero$ Hence, by Limit Comparison test, both $\sum u_n$ *and* $\sum v_n$ converge or diverge together. But $\sum v_n=\sum\frac{1}{n^3}$ **is a $p-series$ with $p=3>1$, so it is convergent.** $\therefore \sum u_n$ **is also convergent.** **Answer**
23	Test the convergence of the series $\sum_{n=1}^{\infty}\frac{\sqrt{n}}{n^2+1}$ **Solution:** $u_n=\frac{\sqrt{n}}{n^2+1}$ Take $v_n=\frac{\sqrt{n}}{n^2}=\frac{1}{n^{2-(\frac{1}{2})}}=\frac{1}{n^{3/2}}$ $\lim_{n\to\infty}\frac{u_n}{v_n}=\lim_{n\to\infty}\frac{\sqrt{n}}{n^2+1}\times n^{3/2}=\lim_{n\to\infty}\frac{1}{1+\frac{1}{n^2}}=1, finite\ and\ non-zero$ Hence, by Limit Comparison test, both $\sum u_n$ *and* $\sum v_n$ converge or diverge together. But $\sum v_n=\sum\frac{1}{n^{3/2}}$ **is a $p-series$ with $p=\frac{3}{2}>1$, so it is convergent.** $\therefore \sum u_n$ **is also convergent.** **Answer**
24	Determine convergence or divergence of series $\sum_{n=1}^{\infty}\frac{(2n^2-1)^{\frac{1}{3}}}{(3n^3+2n+5)^{\frac{1}{4}}}$ **Solution:** $u_n=\frac{(2n^2-1)^{\frac{1}{3}}}{(3n^3+2n+5)^{\frac{1}{4}}}$ Take $v_n=\frac{n^{2/3}}{n^{3/4}}=\frac{1}{n^{(\frac{3}{4})-(\frac{2}{3})}}=\frac{1}{n^{1/12}}$ $\lim_{n\to\infty}\frac{u_n}{v_n}=\lim_{n\to\infty}\frac{(2n^2-1)^{\frac{1}{3}}}{(3n^3+2n+5)^{\frac{1}{4}}}\times n^{1/12}=\lim_{n\to\infty}\frac{(2-\frac{1}{n^2})^{\frac{1}{3}}}{\left(3+\frac{2}{n^2}+\frac{5}{n^3}\right)^{\frac{1}{4}}}$ $=\frac{2^{1/3}}{3^{1/4}}, finite\ and\ non-zero$ Hence, by Limit Comparison test, both $\sum u_n$ *and* $\sum v_n$ converge or diverge together.

	But $\sum v_n = \sum \frac{1}{n^{1/12}}$ **is a** $p-series$ **with** $p = \frac{1}{12} < 1$, **so it is divergent.** $\therefore \sum u_n$ **is also divergent.** **Answer**
25	Find the sum of the series $\sum_{n\geq 1} \frac{4}{(4n-3)(4n+1)}$ **Solution:** $u_n = \frac{4}{(4n-3)(4n+1)}$ Take $v_n = \frac{1}{n^2}$ $\lim_{n\to\infty} \frac{u_n}{v_n} = \lim_{n\to\infty} \frac{4}{(4n-3)(4n+1)} \times n^2 = \lim_{n\to\infty} \frac{4}{\left(4-\frac{3}{n}\right)\left(4+\frac{1}{n}\right)} = 4,$ $= 4, finite\ and\ non-zero$ Hence, by Limit Comparison test, both $\sum u_n$ *and* $\sum v_n$ converge or diverge together. But $\sum v_n = \sum \frac{1}{n^2}$ **is a** $p-series$ **with** $p = 2 > 1$, **so it is convergent.** $\therefore \sum u_n$ **is also convergent.** **Answer**
26	Test the convergence of $\frac{1}{1\cdot 2\cdot 3} + \frac{3}{2\cdot 3\cdot 4} + \frac{5}{3\cdot 4\cdot 5} + \ldots\ldots$ **Solution:** Here, $u_n = \frac{1}{1.2.3} + \frac{3}{2.3.4} + \frac{5}{3.4.5} + \cdots$ $\therefore u_n = \frac{(2n-1)}{n(n+1)(n+2)}$ Take $v_n = \frac{n}{n^3} = \frac{1}{n^2}$ $\lim_{n\to\infty} \frac{u_n}{v_n} = \lim_{n\to\infty} \frac{(2n-1)\times n^2}{n(n+1)(n+2)} = \lim_{n\to\infty} \frac{\left(2-\frac{1}{n}\right)}{\left(1+\frac{1}{n}\right)\left(1+\frac{2}{n}\right)} = \frac{2-0}{(1+0)(1+0)}$ $= 2, finite\ and\ non-zero$ Hence, by Limit Comparison test, both $\sum u_n$ *and* $\sum v_n$ converge or diverge together. But $\sum v_n = \sum \frac{1}{n^2}$ **is a** $p-series$ **with** $p = 2 > 1$, **so it is convergent.** $\therefore \sum u_n$ **is also convergent.** **Answer**
27	Test the convergence of $\frac{3}{1^2-3} + \frac{3}{2^2-3} + \frac{3}{3^2-3} + \frac{3}{4^2-3} \ldots$

	Solution: Here, $u_n = \frac{3}{n^2-3}$ Take $v_n = \frac{1}{n^2}$ $\lim_{n\to\infty} \frac{u_n}{v_n} = \lim_{n\to\infty} \frac{3\times n^2}{n^2-3} = \lim_{n\to\infty} \frac{3}{1-\frac{3}{n^2}} = \frac{3}{1-0} = 3, finite\ and\ non-zero$ Hence, by Limit Comparison test, both $\sum u_n$ *and* $\sum v_n$ converge or diverge together. But $\sum v_n = \sum \frac{1}{n^2}$ **is a $p-series$ with $p = 2 > 1$, so it is convergent.** $\therefore \sum u_n$ **is also convergent.** **Answer**
28	Check for the convergence of the following: $\sum_{n=1}^{\infty} \frac{1}{1^2+2^2+\cdots+n^2}$ **Solution:** Here, $u_n = \frac{1}{1^2+2^2+\cdots+n^2} = \frac{1}{\sum n^2} = \frac{6}{n(n+1)(2n+1)}$ Take $v_n = \frac{1}{n^3}$ $\lim_{n\to\infty} \frac{u_n}{v_n} = \lim_{n\to\infty} \frac{6\times n^3}{n(n+1)(2n+1)} = \lim_{n\to\infty} \frac{6}{\left(1+\frac{1}{n}\right)\left(2+\frac{1}{n}\right)} = 6, finite\ and\ non-zero$ Hence, by Limit Comparison test, both $\sum u_n$ *and* $\sum v_n$ converge or diverge together. But $\sum v_n = \sum \frac{1}{n^3}$ **is a $p-series$ with $p = 3 > 1$, so it is convergent.** $\therefore \sum u_n$ **is also convergent.** **Answer**
29	Test the convergence of the series $\frac{1.2}{3^2.4^2} + \frac{3.4}{5^2 6^2} + \frac{5.6}{7^2 8^2} + \cdots$ **Solution:** Here, $u_n = \frac{1.2}{3^2.4^2} + \frac{3.4}{5^2 6^2} + \frac{5.6}{7^2 8^2} + \cdots$ $\therefore u_n = \frac{(2n-1)(2n)}{(2n+1)^2(2n+2)^2}$ Take $v_n = \frac{1}{n^2}$

$$\lim_{n\to\infty}\frac{u_n}{v_n}=\lim_{n\to\infty}\frac{(2n-1)(2n)\times n^2}{(2n+1)^2(2n+2)^2}=\lim_{n\to\infty}\frac{\left(2-\frac{1}{n}\right)(2)}{\left(2+\frac{1}{n}\right)^2\left(2+\frac{2}{n}\right)^2}$$

$$=2, finite\ \ and\ non-zero$$

Hence, by Limit Comparison test, both $\sum u_n$ *and* $\sum v_n$ converge or diverge together.

But $\sum v_n=\sum\frac{1}{n^2}$ **is a** $p-series$ **with** $p=2>1$, **so it is convergent.**

$\therefore \sum u_n$ **is also convergent.** **Answer**

TOPIC:8 RATIO TEST

RATIO TEST:

Let $\sum u_n$ be a positive term series such that

$$\lim_{n\to\infty}\frac{u_{n+1}}{u_n}=l \text{ then,}$$

(1) $\sum u_n$ converges if $l<1$
(2) $\sum u_n$ diverges if $l>1$
(3) The test fails if $l=1$

RAABE'S TEST:

Let $\sum u_n$ be a positive term series such that

$$\lim_{n\to\infty} n\left(\frac{u_n}{u_{n+1}}-1\right)=l \text{ then,}$$

- $\sum u_n$ converges if $l>1$
- $\sum u_n$ diverges if $l<1$
- The test fails if $l=1$

NUMERICAL QUESTIONS

1 Test the convergence and Divergence of the following series $\sum_{n=1}^{\infty}\frac{n^3+2}{2^n+2}$

Solution:

$$u_n=\frac{n^3+2}{2^n+2}$$

$$u_{n+1}=\frac{(n+1)^3}{2^{n+1}+2}$$

	$\lim_{n\to\infty}\frac{u_{n+1}}{u_n}=\lim_{n\to\infty}\frac{(n+1)^3}{2^{n+1}+2}\times\frac{2^n+2}{n^3+2}=\lim_{n\to\infty}\frac{n^3\left(1+\frac{1}{n}\right)^3}{2^n\left(2+\frac{2}{2^n}\right)}\times\frac{2^n\left(1+\frac{2}{2^n}\right)}{n^3\left(1+\frac{2}{n^3}\right)}=\frac{(1+0)(1+0)}{(2+0)(1+0)}$ $=\frac{1}{2}<1$ $\therefore$ **By Ratio test $\sum u_n$ is convergent.** **Answer**
2	Test the convergence of $\sum_{n=1}^{\infty}\frac{2^n}{n^3+1}$ **Solution:** $u_n=\frac{2^n}{n^3+1}$ $u_{n+1}=\frac{2^{n+1}}{(n+1)^3+1}n^3$ $\lim_{n\to\infty}\frac{u_{n+1}}{u_n}=\lim_{n\to\infty}\frac{2^{n+1}}{(n+1)^3+1}\times\frac{n^3+1}{2^n}=\lim_{n\to\infty}\frac{2^n(2)}{n^3\left[\left(1+\frac{1}{n}\right)^3+\frac{1}{n^3}\right]}\times\frac{n^3\left(1+\frac{1}{n^3}\right)}{2^n}$ $=\frac{2(1+0)}{(1+0)}=2>1$ $\therefore$ **By Ratio test $\sum u_n$ is divergent.** **Answer**
3	Determine the convergence of $\sum_{n=1}^{\infty}\frac{1}{5^n-1}$ **Solution:** $u_n=\frac{1}{5^n-1}$ Take $u_{n+1}=\frac{1}{5^{n+1}-1}$ $\lim_{n\to\infty}\frac{u_{n+1}}{u_n}=\lim_{n\to\infty}\left(\frac{1}{5^{n+1}-1}\right)\times(5^n-1)=\lim_{n\to\infty}\frac{5^n}{5^n}\left(\frac{1-\frac{1}{5^n}}{5-\frac{1}{5^n}}\right)$ $=\frac{1}{5}<1$ $\therefore$ **By Ratio test, $\sum u_n$ is convergent.** **Answer**
4	Test the convergence of $\sum_{n=1}^{\infty}\frac{2^n+1}{3^n+1}$ **Solution:**

	$u_n = \frac{2^n+1}{3^n+1}$ Take $u_{n+1} = \frac{2^{n+1}+1}{3^{n+1}+1}$ $\lim_{n\to\infty} \frac{u_{n+1}}{u_n} = \lim_{n\to\infty} \left(\frac{2^{n+1}+1}{3^{n+1}+1}\right) \times \left(\frac{3^n+1}{2^n+1}\right) = \lim_{n\to\infty} \frac{2^n}{3^n}\left(\frac{2+\frac{1}{2^n}}{3+\frac{1}{3^n}}\right) \times \frac{3^n}{2^n}\left(\frac{1+\frac{1}{3^n}}{1+\frac{1}{2^n}}\right)$ $= \frac{2}{3} < 1$ ∴ **By Ratio test, $\sum u_n$ is convergent.** **Answer**
5	Test the convergence of $\sum_{n=1}^{\infty} \frac{n^2}{3^n}$ **Solution:** $u_n = \frac{n^2}{3^n}$ $u_{n+1} = \frac{(n+1)^2}{3^{n+1}}$ $\lim_{n\to\infty} \frac{u_{n+1}}{u_n} = \lim_{n\to\infty} \frac{(n+1)^2}{3^{n+1}} \times \frac{3^n}{n^2} = \lim_{n\to\infty} \frac{n^2\left(1+\frac{1}{n}\right)^2}{3^n(3)} \times \frac{3^n}{n^2} = \frac{(1+0)}{(3)}$ $= \frac{1}{3} < 1$ ∴ **By Ratio test $\sum u_n$ is convergent.** **Answer**
6	Discuss the convergence of $\sum_{n=1}^{\infty} \frac{n^2}{2^n}$ **Solution:** $u_n = \frac{n^2}{2^n}$ $u_{n+1} = \frac{(n+1)^2}{2^{n+1}}$ $\lim_{n\to\infty} \frac{u_{n+1}}{u_n} = \lim_{n\to\infty} \frac{(n+1)^2}{2^{n+1}} \times \frac{2^n}{n^2} = \lim_{n\to\infty} \frac{n^2\left(1+\frac{1}{n}\right)^2}{2^n(2)} \times \frac{2^n}{n^2} = \frac{(1+0)}{(2)}$ $= \frac{1}{2} < 1$ ∴ **By Ratio test $\sum u_n$ is convergent.** **Answer**

7	Test for convergence or divergence: $\sum_{n=1}^{\infty} \frac{n^{10}}{10^n}$ **Solution:** $u_n = \frac{n^{10}}{10^n}$ $u_{n+1} = \frac{(n+1)^{10}}{10^{n+1}}$ $\lim_{n\to\infty} \frac{u_{n+1}}{u_n} = \lim_{n\to\infty} \frac{(n+1)^{10}}{10^{n+1}} \times \frac{10^n}{n^2} = \lim_{n\to\infty} \frac{n^{10}\left(1+\frac{1}{n}\right)^{10}}{10^n(10)} \times \frac{10^n}{n^{10}} = \frac{(1+0)^{10}}{(10)}$ $= \frac{1}{10} < 1$ $\therefore$ **By Ratio test $\sum u_n$ is convergent.** **Answer**
8	Discuss the convergence of the following series $\sum_{n=1}^{\infty} n!\, e^{-n}$ **Solution:** $u_n = n!\, e^{-n}$ Take $u_{n+1} = (n+1)!\, e^{-(n+1)}$ $\lim_{n\to\infty} \frac{u_{n+1}}{u_n} = \lim_{n\to\infty} \frac{\left((n+1)!\, e^{-(n+1)}\right)}{n!\, e^{-n}} = \lim_{n\to\infty} \left(\frac{(n+1)n!e^{-n}}{n!e^{-n}}\right) \times \frac{1}{e} \to \infty$ $\therefore$ **By Ratio test, $\sum u_n$ is divergent.** **Answer**
9	Test the D'Alembert's ratio test. Discuss the convergence of the following series $\sum_{n=1}^{\infty} \frac{4^n(n+1)!}{n^{n+1}}$ **Solution:** $u_n = \frac{4^n(n+1)!}{n^{n+1}}$ Take $u_{n+1} = \frac{4^{n+1}\,(n+2)!}{(n+1)^{n+2}}$ $\lim_{n\to\infty} \frac{u_{n+1}}{u_n} = \lim_{n\to\infty} \frac{4^{n+1}\,(n+2)!}{(n+1)^{n+2}} \times \frac{n^{n+1}}{4^n(n+1)!} = \lim_{n\to\infty} \left(\frac{4^n\, 4\,(n+2)(n+1)!}{(n+1)^n(n+1)^2}\right) \times \frac{n^n\, n}{4^n(n+1)!}$ $= \frac{4(1+0)}{(1+0)^2 \lim_{n\to\infty}\left(1+\frac{1}{n}\right)^n} = \frac{4}{e} > 1 \;\; (e = 2.718)$ $\therefore$ **By Ratio test, $\sum u_n$ is divergent.** **Answer**

10	Test the convergence of the series $\sum_{n=1}^{\infty} \frac{n!}{n^n}$ **Solution:** $u_n = \frac{(n)!}{n^n}$ Take $u_{n+1} = \frac{(n+1)!}{(n+1)^{n+1}}$ $\lim_{n\to\infty} \frac{u_{n+1}}{u_n} = \lim_{n\to\infty} \frac{(n+1)!}{(n+1)^{n+1}} \times \frac{n^n}{(n)!} = \lim_{n\to\infty} \left(\frac{(n+1)(n)!}{(n+1)^n(n+1)^1}\right) \times \frac{n^n}{(n)!}$ $= \lim_{n\to\infty} \frac{1}{\left(1+\frac{1}{n}\right)^n} = \frac{1}{e} < 1$ $\therefore$ **By Ratio test, $\sum u_n$ is convergent.** **Answer**
11	Investigate the convergence of $\sum_{n=1}^{\infty} \frac{2^n}{n!}$. **Solution:** $u_n = \frac{2^n}{(n)!}$ Take $u_{n+1} = \frac{2^{n+1}}{(n+1)!}$ $\lim_{n\to\infty} \frac{u_{n+1}}{u_n} = \lim_{n\to\infty} \frac{2^{n+1}}{(n+1)!} \times \frac{(n)!}{2^n} = \lim_{n\to\infty} \left(\frac{2^n\, 2}{(n+1)(n)!}\right) \times \frac{(n)!}{2^n!}$ $= \lim_{n\to\infty} \frac{2}{(n+1)} = 0 < 1$ $\therefore$ **By Ratio test, $\sum u_n$ is convergent.** **Answer**
12	Test the convergence of $\sum_{n=1}^{\infty} \frac{3^n\, n!}{n^n}$ by Ratio Test. **Solution:** $u_n = \frac{3^n n!}{n^n}$ Take $u_{n+1} = \frac{3^{n+1}\,(n+1)!}{(n+1)^{n+1}}$

	$\lim_{n\to\infty}\frac{u_{n+1}}{u_n}=\lim_{n\to\infty}\frac{3^{n+1}(n+1)!}{(n+1)^{n+1}}\times\frac{(n^n)}{3^n n!}=\lim_{n\to\infty}\left(\frac{3^n\,3\,(n+1)(n!)}{(n+1)^n(n+1)^1}\right)\times\frac{(n^n)}{3^n n!}$ $=\lim_{n\to\infty}\frac{3}{\left(1+\frac{1}{n}\right)^n}=\frac{3}{e}>1$ ∴ **By Ratio test, $\sum u_n$ is divergent.** **Answer**
13	Test the Convergence for the series $\sum_{n=1}^{\infty}\frac{4^n n!n!}{(2n)!}$ **Solution:** $u_n=\frac{4^n n!\,n!}{(2n)!}$ Take $u_{n+1}=\frac{4^{n+1}(n+1)!(n+1)!}{(2n+2)!}$ $\lim_{n\to\infty}\frac{u_{n+1}}{u_n}=\lim_{n\to\infty}\frac{4^{n+1}(n+1)!\,(n+1)!}{(2n+2)!}\times\frac{(2n)!}{4^n n!\;n!}$ $=\lim_{n\to\infty}\frac{4^n\,4\,(n+1)\,n!\,(n+1)\,n!}{(2n+2)(2n+1)(2n)!}\times\frac{(2n)!}{4^n n!\,n!}$ $=\lim_{n\to\infty}\frac{4(n+1)^2}{2(n+1)(2n+1)}=\lim_{n\to\infty}\frac{2(n+1)}{(2n+1)}=1$ ∴ **Ratio test fails** **We can not conclude whether $\sum u_n$ is convergent or divergent.** Using Raabe's test, $\lim_{n\to\infty}n\left(\frac{u_n}{u_{n+1}}-1\right)=(-1/2)<1\Rightarrow Convergent$
14	Test the convergence and Divergence of the following series $\sum_{n=1}^{\infty}\frac{n2^n(n+1)!}{3^n n!}$ **Solution:** $u_n=\frac{n2^n(n+1)!}{3^n n!}$ Take $u_{n+1}=\frac{(n+1)2^{n+1}(n+2)!}{3^{n+1}(n+1)!}$

	$\lim_{n\to\infty}\frac{u_{n+1}}{u_n}=\lim_{n\to\infty}\frac{(n+1)2^{n+1}(n+2)!}{3^{n+1}(n+1)!}\times\frac{3^n n!}{n2^n(n+1)!}$ $=\lim_{n\to\infty}\left(\frac{(n+1)2^n\,2\,(n+2)(n+1)!}{3^n\,3\,(n+1)\,n!}\right)\times\frac{(3^n n!)}{n2^n(n+1)!}$ $=\lim_{n\to\infty}\frac{2}{3}\left(1+\frac{2}{n}\right)=\frac{2}{3}<1$ $\therefore$ **By Ratio test, $\sum u_n$ is convergent.** **Answer**
15	Determine whether the series $\sum_{n=1}^{\infty}\frac{(n-1)!}{n^2}$ converges or diverges. **Solution:** $u_n=\frac{(n-1)!}{n^2}$ Take $u_{n+1}=\frac{(n)!}{(n+1)^2}$ $\lim_{n\to\infty}\frac{u_{n+1}}{u_n}=\lim_{n\to\infty}\frac{(n)!}{(n+1)^2}\times\frac{n^2}{(n-1)!}=\lim_{n\to\infty}\left(\frac{(n)(n-1)!}{n^2(1+\frac{1}{n})^2}\right)\times\frac{n^2}{(n-1)!}$ $=\lim_{n\to\infty}\frac{n}{(1+\frac{1}{n})^2}\to\infty>1$ $\therefore$ **By Ratio test, $\sum u_n$ is divergent.** **Answer**
16	Test the convergence or divergence of the series $\sum_{n=1}^{\infty}\frac{1}{(n)!}$ **Solution:** $u_n=\frac{1}{(n)!}$ Take $u_{n+1}=\frac{1}{(n+1)!}$ $\lim_{n\to\infty}\frac{u_{n+1}}{u_n}=\lim_{n\to\infty}\frac{1}{(n+1)!}\times\frac{(n)!}{1}=\lim_{n\to\infty}\left(\frac{1}{(n+1)(n)!}\right)\times\frac{(n)!}{1}$ $=\lim_{n\to\infty}\frac{1}{(n+1)}=0<1$ $\therefore$ **By Ratio test, $\sum u_n$ is convergent.** **Answer**
17	Test the convergence of the series $1+\frac{2^2}{2!}+\frac{3^2}{3!}+\frac{4^2}{4!}+\cdots$

	Solution: $u_n = \frac{n^2}{(n)!}$ Take $u_{n+1} = \frac{(n+1)^2}{(n+1)!}$ $\lim_{n\to\infty} \frac{u_{n+1}}{u_n} = \lim_{n\to\infty} \frac{(n+1)^2}{(n+1)!} \times \frac{(n)!}{n^2} = \lim_{n\to\infty} \left(\frac{(n+1)^2}{(n+1)(n)!}\right) \times \frac{(n)!}{n^2}$ $= \lim_{n\to\infty} \frac{(n+1)}{n^2} = \lim_{n\to\infty} \left(\frac{1}{n} + \frac{1}{n^2}\right) = 0 < 1$ $\therefore$ **By Ratio test, $\sum u_n$ is convergent.** **Answer**
18	Test for the convergence for the series $\frac{1}{2!} + \frac{2}{3!} + \frac{3}{4!} + \frac{4}{5!} + \ldots$ **Solution:** $u_n = \frac{n}{(n+1)!}$ Take $u_{n+1} = \frac{n+1}{(n+2)!}$ $\lim_{n\to\infty} \frac{u_{n+1}}{u_n} = \lim_{n\to\infty} \frac{n+1}{(n+2)!} \times \frac{(n+1)!}{n} = \lim_{n\to\infty} \left(\frac{n+1}{(n+2)(n+1)!}\right) \times \frac{(n+1)!}{n}$ $= \lim_{n\to\infty} \left(\frac{1+\frac{1}{n}}{1+\frac{2}{n}}\right) \times \frac{1}{n} = 1 \times 0 = 0 < 1$ $\therefore$ **By Ratio test, $\sum u_n$ is convergent.** **Answer**
19	Discuss the convergence of $\frac{1}{2\sqrt{1}} + \frac{x^2}{3\sqrt{2}} + \frac{x^4}{4\sqrt{3}} + \frac{x^6}{5\sqrt{4}} + \ldots$ **Solution:** Here, $u_n = \frac{x^{2n-2}}{(n+1)\sqrt{n}}$ $\therefore u_{n+1} = \frac{x^{2n}}{(n+2)\sqrt{n+1}}$ $\lim_{n\to\infty} \frac{u_{n+1}}{u_n} = \lim_{n\to\infty} \frac{x^{2n}}{(n+2)\sqrt{n+1}} \times \frac{(n+1)\sqrt{n}}{x^{2n-2}} = x^2$ $\therefore$**By Ratio test,** (1) $\sum u_n$ converges if $l = x^2 < 1$ (2) $\sum u_n$ diverges if $l = x^2 > 1$

(3) The test fails if $l = x^2 = 1$

When $x^2 = 1$,

$$u_n = \frac{x^{2n-2}}{(n+1)\sqrt{n}} = \frac{\frac{x^{2n}}{x^2}}{(n+1)\sqrt{n}} = \frac{1}{(n+1)\sqrt{n}}$$

Take $v_n = \frac{1}{n^{3/2}}$

$$\lim_{n\to\infty} \frac{u_n}{v_n} = \lim_{n\to\infty} \left(\frac{1}{(n+1)\sqrt{n}}\right) \times (n^{3/2}) = 1, finite\ and\ non-zero$$

Hence, by Limit Comparison test, both $\sum u_n$ *and* $\sum v_n$ converge or diverge together.

But $\sum v_n = \sum \frac{1}{n^{3/2}}$ is convergent as $p = \frac{3}{2} > 1$

$\therefore \sum u_n$ is also convergent for $x^2 = 1$

Hence, $\sum u_n$ is convergent for $x^2 \leq 1$ And is divergent for $x^2 > 1$ Answer

20 Discuss the convergence of the series $\frac{x}{1.2} + \frac{x^2}{2.3} + \frac{x^3}{3.4} + \cdots$, where $x > 0$

Solution:

Here, $u_n = \frac{x^n}{n(n+1)}$

$\therefore u_{n+1} = \frac{x^{n+1}}{(n+1)(n+2)}$

$$\lim_{n\to\infty} \frac{u_{n+1}}{u_n} = \lim_{n\to\infty} \left(\frac{x^{n+1}}{(n+1)(n+2)}\right) \times \left(\frac{n(n+1)}{x^n}\right) = \lim_{n\to\infty} \frac{x}{\left(1+\frac{2}{n}\right)} = x,$$

∴By Ratio test,

(1) $\sum u_n$ converges if $l = x < 1$
(2) $\sum u_n$ diverges if $l = x > 1$
(3) The test fails if $l = x = 1$

When $x = 1$,

$$u_n = \frac{1}{n(n+1)}$$

Take $v_n = \frac{1}{n^2}$

$$\lim_{n\to\infty} \frac{u_n}{v_n} = \lim_{n\to\infty} \left(\frac{1}{n(n+1)}\right) \times (n^2) = 1, finite\ and\ non-zero$$

Hence, by Limit Comparison test, both $\sum u_n$ *and* $\sum v_n$ converge or diverge together.

But $\sum v_n = \sum \frac{1}{n^2}$ is convergent as $p = 2 > 1$

$\therefore \sum u_n$ is also convergent for $x = 1$

Hence, $\sum u_n$ is convergent for $x \leq 1$ And is divergent for $x > 1$ **Answer**

21 Test for convergence the series

$$2+\frac{3}{2}x+\frac{4}{3}x^2+\frac{5}{4}x^3+\cdots$$

Solution:

Here, $u_n = \frac{(n+1)x^{n-1}}{n}$

$\therefore u_{n+1} = \frac{(n+2)\,x^n}{(n+1)}$

$$\lim_{n\to\infty}\frac{u_{n+1}}{u_n} = \lim_{n\to\infty}\left(\frac{(n+2)\,x^n}{(n+1)}\right)\times\left(\frac{n}{(n+1)x^{n-1}}\right) = \lim_{n\to\infty}\frac{x\left(1+\frac{2}{n}\right)}{\left(1+\frac{1}{n}\right)\left(1+\frac{1}{n}\right)} = x,$$

∴By Ratio test,

(1) $\sum u_n$ converges if $l = x < 1$
(2) $\sum u_n$ diverges if $l = x > 1$
(3) The test fails if $l = x = 1$

When $x = 1$,

$$u_n = \frac{(n+1)}{n}$$

$$\lim_{n\to\infty} u_n = \lim_{n\to\infty}\frac{(n+1)}{n} = 1 \neq 0$$

Hence, by Zero test (Divergence test), $\sum \boldsymbol{u_n}$ diverges

$\therefore \sum u_n$ is divergent for $x = 1$

Hence, $\sum u_n$ is convergent for $x < 1$ And is divergent for $x \geq 1$ **Answer**

TOPIC:9 CAUCHY'S ROOT TEST

Exploring Emerging Technologies

Statement of Cauchy's Root test

Let $\sum u_n$ be a positive term series such that

$$\lim_{n\to\infty}\sqrt[n]{u_n} = l \text{ then}$$

- $\sum u_n$ converges if $l < 1$

	• $\sum u_n$ diverges if $l > 1$ • The test fails if $l = 1$
	NUMERICAL QUESTIONS
1	Test the convergence of the following series $\sum_{n=1}^{\infty}\left(1+\frac{1}{n}\right)^{n^2}$ **Solution:** $u_n = \left(1+\frac{1}{n}\right)^{n^2} \Rightarrow (u_n)^{1/n} = \left[\left(1+\frac{1}{n}\right)^{n^2}\right]^{1/n} = \left(1+\frac{1}{n}\right)^{n}$ $\therefore \lim_{n\to\infty}(u_n)^{1/n} = \lim_{n\to\infty}\left(1+\frac{1}{n}\right)^{n} = e > 1$ $\therefore$ **By Cauchy's Root Test,** $\sum u_n$ **is divergent.** **Answer**
2	Determine whether series converges or diverges $\sum_{n=1}^{\infty}\left(\frac{2n+3}{3n+12}\right)^{n}$ **Solution:** $u_n = \left(\frac{2n+3}{3n+12}\right)^{n} \Rightarrow (u_n)^{1/n} = \left[\left(\frac{2n+3}{3n+12}\right)^{n}\right]^{1/n} = \left(\frac{2n+3}{3n+12}\right)$ $\therefore \lim_{n\to\infty}(u_n)^{1/n} = \lim_{n\to\infty}\left(\frac{2n+3}{3n+12}\right) = \frac{2}{3} < 1$ $\therefore$ **By Cauchy's Root Test,** $\sum u_n$ **is convergent.** **Answer**
3	Test Cauchy's root test. Discuss the convergence of the series $\sum_{n=2}^{\infty}\frac{n}{(\log n)^n}$ **Solution:** $u_n = \frac{n}{(\log n)^n} \Rightarrow (u_n)^{1/n} = \frac{n^{1/n}}{\log n}$ $\therefore \lim_{n\to\infty}(u_n)^{1/n} = \lim_{n\to\infty}\left(\frac{1}{\log n}\right) \left(\because \lim_{n\to\infty} n^{1/n} = 1\right)$ $= 0 < 1$ $\therefore$ ***The given series is convergent.*** **By Cauchy's Root Test,**
4	Test the convergence of the series $\sum_{n=1}^{\infty}\left(\frac{1}{1+n}\right)^{n}$ **Solution:**

	$u_n = \left(\frac{1}{1+n}\right)^n \Rightarrow (u_n)^{1/n} = \left[\left(\frac{1}{1+n}\right)^n\right]^{1/n} = \left(\frac{1}{1+n}\right)$ $\therefore \lim_{n\to\infty}(u_n)^{1/n} = \lim_{n\to\infty}\left(\frac{1}{1+n}\right) = 0 < 1$ $\therefore$ **By Cauchy's Root Test, $\sum u_n$ is convergent.** **Answer**
5	Test the convergence of the series $\frac{1}{3} + \left(\frac{2}{5}\right)^2 + \left(\frac{3}{7}\right)^3 + \cdots + \left(\frac{n}{2n+1}\right)^n + \cdots$ **Solution:** $u_n = \left(\frac{n}{2n+1}\right)^n \Rightarrow (u_n)^{1/n} = \left[\left(\frac{n}{2n+1}\right)^n\right]^{1/n} = \left(\frac{n}{2n+1}\right)$ $\therefore \lim_{n\to\infty}(u_n)^{1/n} = \lim_{n\to\infty}\left(\frac{n}{2n+1}\right) = \lim_{n\to\infty}\left(\frac{1}{2+\frac{1}{n}}\right) = \frac{1}{2} < 1$ $\therefore$ **By Cauchy's Root Test, $\sum u_n$ is convergent.** **Answer**
6	Test the convergence of the series $\sum_{n=1}^{\infty} \frac{n}{e^{-n}}$ **Solution:** $u_n = \frac{n}{e^{-n}} \Rightarrow (u_n)^{1/n} = \left[\frac{n}{e^{-n}}\right]^{1/n} = \left(\frac{n^{1/n}}{e^{-1}}\right)$ $\therefore \lim_{n\to\infty}(u_n)^{1/n} = \lim_{n\to\infty}\left(\frac{n^{1/n}}{e^{-1}}\right) = e > 1$ $\therefore$ **By Cauchy's Root Test, $\sum u_n$ is divergent.** **Answer**
7	Test for convergence the series whose nth term is $\left(1+\frac{1}{\sqrt{n}}\right)^{n^{(-\frac{3}{2})}}$ **Solution:** $u_n = \left(1+\frac{1}{\sqrt{n}}\right)^{n^{(-\frac{3}{2})}} \Rightarrow (u_n)^{1/n} = \left[\left(1+\frac{1}{\sqrt{n}}\right)^{n^{(-\frac{3}{2})}}\right]^{1/n}$ $= \left(1+\frac{1}{\sqrt{n}}\right)^{n^{\frac{-3}{2}}\times n^{-1}} = \left(1+\frac{1}{\sqrt{n}}\right)^{n^{\frac{-5}{2}}} = \left[\left(1+\frac{1}{\sqrt{n}}\right)^{\sqrt{n}}\right]^{-5}$

	$\therefore \lim_{n\to\infty}(u_n)^{1/n} = \lim_{n\to\infty}\left[\left(1+\frac{1}{\sqrt{n}}\right)^{\sqrt{n}}\right]^{-5} = \frac{1}{e^5} < 1$ $\therefore$ **By Cauchy's Root Test, $\sum u_n$ is convergent.** **Answer**
8	Test the convergence or divergence of the series $\sum_{n=1}^{\infty}\frac{1}{\left(1+\frac{1}{n}\right)^{n^2}}$ **Solution:** $u_n = \frac{1}{\left(1+\frac{1}{n}\right)^{n^2}} \Rightarrow (u_n)^{1/n} = \left[\frac{1}{\left(1+\frac{1}{n}\right)^{n^2}}\right]^{1/n} = \frac{1}{\left(1+\frac{1}{n}\right)^{n}}$ $\therefore \lim_{n\to\infty}(u_n)^{1/n} = \lim_{n\to\infty}\frac{1}{\left(1+\frac{1}{n}\right)^{n}} = \frac{1}{e} < 1$ $\therefore$ **By Cauchy's Root Test, $\sum u_n$ is convergent.** **Answer**
9	Test the convergence of the following series $\sum_{n=1}^{\infty}\left(\frac{n+1}{n+2}\right)^{n} x^n \,;\, x > 0$ **Solution:** **Statement of Cauchy's root test** Let $\sum u_n$ be a positive series such that $\lim_{n\to\infty}\sqrt[n]{u_n} = l$ then • $\sum u_n$ converges if $l < 1$ • $\sum u_n$ diverges if $l > 1$ • The test fails if $l = 1$ $u_n = \left(\frac{n+1}{n+2}\right)^{n} x^n \Rightarrow (u_n)^{\frac{1}{n}} = \left[\left(\frac{n+1}{n+2}\right)^{n} x^n\right]^{\frac{1}{n}} = \left(\frac{n+1}{n+2}\right) x$ $\lim_{n\to\infty}\sqrt[n]{u_n} = \lim_{n\to\infty}\left(\frac{n+1}{n+2}\right) x = x$ $\therefore$**By Cauchy's Root test,** (1) $\sum u_n$ converges if $l = x < 1$ (2) $\sum u_n$ diverges if $l = x > 1$

	(3) The test fails if $l = x = 1$ **When $x = 1$,** $u_n = \left(\frac{n+1}{n+2}\right)^n$ $\lim_{n\to\infty} u_n = \lim_{n\to\infty} \left(\frac{n+1}{n+2}\right)^n = \lim_{n\to\infty} u_n \frac{\left(1+\frac{1}{n}\right)^n}{\left[\left(1+\frac{2}{n}\right)^{\frac{n}{2}}\right]^2} = \frac{e}{e^2} = \frac{1}{e} \neq 0$ $\therefore$ By Zero Test(Divergence Test), $\sum u_n$ diverges when $x = 1$ **Thus, $\sum u_n$ converges if $l = x < 1$ and diverges if $l = x \geq 1$ Answer**
10	Test the convergence of the series $\sum_{n=1}^{\infty} \frac{n^n x^n}{(n+1)^n}$, $x > 0$ **Solution:** $u_n = \left(\frac{n}{n+1}\right)^n x^n \Rightarrow (u_n)^{\frac{1}{n}} = \left[\left(\frac{n}{n+1}\right)^n x^n\right]^{\frac{1}{n}} = \left(\frac{n}{n+1}\right) x$ $\lim_{n\to\infty} \sqrt[n]{u_n} = \lim_{n\to\infty} \left(\frac{n}{n+1}\right) x = x$ **$\therefore$By Cauchy's Root test,** **(1) $\sum u_n$ converges if $l = x < 1$** **(2) $\sum u_n$ diverges if $l = x > 1$** **(3) The test fails if $l = x = 1$** **When $x = 1$,** $u_n = \left(\frac{n}{n+1}\right)^n$ $\lim_{n\to\infty} u_n = \lim_{n\to\infty} \left(\frac{n}{n+1}\right)^n = \lim_{n\to\infty} u_n \frac{1}{\left(1+\frac{1}{n}\right)^n} = \frac{1}{e} \neq 0$ $\therefore$ By Zero Test(Divergence Test), $\sum u_n$ diverges when $x = 1$ **Thus, $\sum u_n$ converges if $l = x < 1$ and diverges if $l = x \geq 1$ Answer**
11	Test the convergence of $\sum_{n=1}^{\infty} \frac{[(n+1)x]^n}{n^{n+1}}$ **Solution** Let $u_n = \frac{(n+1)^n x^n}{n^{n+1}}$

$$(u_n)^{\frac{1}{n}} = \frac{(n+1)x}{n^{\frac{n+1}{n}}}$$

$$= \frac{(n+1)x}{n.n^{\frac{1}{n}}}$$

$$\lim_{n\to\infty} (u_n)^{\frac{1}{n}} = \lim_{n\to\infty} \left(1+\frac{1}{n}\right)\frac{x}{n^{\frac{1}{n}}}$$

$$= x \qquad \left[\because \lim_{n\to\infty} n^{\frac{1}{n}} = 1\right]$$

Hence, by **Cauchy's Root test,**

(1)The series is convergent, if $x < 1$ and

(2) Divergent if $x > 1$.

(3)The test fails for $x = 1$.

If $x = 1$, then $\quad u_n = \frac{(n+1)^n}{n^{n+1}}$

$$= \frac{(n+1)^n}{n.n^n}$$

$$= \frac{1}{n}\left(\frac{n+1}{n}\right)^n$$

$$= \frac{1}{n}\left(1+\frac{1}{n}\right)^n$$

Let $\quad v_n = \frac{1}{n}$

$$\lim_{n\to\infty} \frac{u_n}{v_n} = \lim_{n\to\infty} \left(1+\frac{1}{n}\right)^n$$

$$= e \qquad \text{[finite and non-zero]}$$

Hence, by **Limit Comparison test**, both $\sum u_n$ *and* $\sum v_n$ converge or diverge together.

But $\sum v_n = \sum \frac{1}{n}$ **is divergent as p=1**.

$\sum u_n$ is also divergent for $x = 1$,

Hence, the series is convergent for $x < 1$ and divergent for $x \geq 1$.

TOPIC:10 ALTERNATING SERIES TEST (LEIBNITZ'S TEST), ABSOLUTE AND CONDITIONAL CONVERGENCE

- An infinite series in which terms are alternatively positive and negative is called an alternating series.

Leibnitz's Test for Alternating Series

Statement:

The alternating series $u_1 - u_2 + u_3 - u_4 \ldots$ is convergent if

(1) $|u_n| > |u_{n+1}|$

(2) $\lim_{n\to\infty} |u_n| = \lim_{n\to\infty} \frac{1}{n^2} = 0$

If $\lim_{n\to\infty} |u_n| \neq 0$ then $\sum u_n$ is known as an Oscillatory series.

Absolute and Conditional Convergence of a Series:

A series $\sum u_n$ converges absolutely if the corresponding series $\sum |u_n|$ converges.

Conditionally Convergent:

A series $\sum u_n$ converges Conditionally if

- $\sum u_n$ converges but
- $\sum |u_n|$ diverges

NOTE: Every absolutely convergent series is a convergent series.

But the converse is not true.

SHORT QUESTIONS

1 The series $\sum_{n=1}^{\infty} \frac{(-1)^n}{n!}$ Is

Solution:

$|u_n| = \left|\frac{(-1)^n}{n!}\right| = \frac{1}{n!}$

$|u_{n+1}| = \frac{1}{(n+1)!}$

$\lim_{n\to\infty} \frac{|u_{n+1}|}{|u_n|} = \lim_{n\to\infty} \frac{n!}{(n+1)!} = \lim_{n\to\infty} \frac{n!}{(n+1)n!} = 0 < 1$

$\therefore \sum u_n$ is absolutely convergent and hence convergent. **Answer**

2 The sum of the series $1 - \frac{1}{2} + \frac{1}{2^2} - \frac{1}{2^3} + \ldots$ is ___

	Solution: $\lvert u_n \rvert = \left\lvert \frac{(-1)^n}{(2)^n} \right\rvert$ $= \frac{1}{(2)^n}$ $\sqrt[n]{\lvert u_n \rvert} = \frac{1}{2}$ $\lim_{n\to\infty} \sqrt[n]{\lvert u_n \rvert} = \frac{1}{2} < 1$ $\therefore \sum u_n$ converges absolutely and hence converges to $\frac{1}{2}$. **Answer**
	NUMERICAL QUESTIONS
	TEST THE CONVERGENCE AND DIVERGENCE
1	Test the convergence and Divergence of the following series $\sum_{n=1}^{\infty} \frac{(-1)^n (n+1)^n}{(2n)^n}$ **Solution:** The given series is an alternating series $\lvert u_n \rvert = \left\lvert \frac{(-1)^n (n+1)^n}{(2n)^n} \right\rvert$ $= \frac{(n+1)^n}{(2n)^n}$ $\sqrt[n]{\lvert u_n \rvert} = \frac{n+1}{2n} = (\frac{1}{2} + \frac{1}{2n})$ $\lim_{n\to\infty} \sqrt[n]{\lvert u_n \rvert} = \frac{1}{2} < 1$ $\therefore \sum u_n$ is absolutely convergent and hence convergent. **Answer**
2	Check the convergence of $\sum_{n=1}^{\infty} (-1)^{n+1} \frac{1}{n^2}$ Test the convergence of the series $\frac{1}{1^2} - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \cdots$ **Solution:** The given series is an alternating series $u_n = \frac{(-1)^{n+1}}{n^2}$ $\lvert u_n \rvert = \left\lvert \frac{(-1)^{n+1}}{n^2} \right\rvert = \frac{1}{n^2}$ $\therefore \lvert u_{n+1} \rvert = \frac{1}{(n+1)^2} = \frac{1}{n^2+2n+1}$

	(1) $\lvert u_n\rvert - \lvert u_{n+1}\rvert = \frac{1}{n^2} - \frac{1}{n^2+2n+1} = \frac{2n+1}{(n^2)(n^2+2n+1)} > 0, for\ all\ n \in N$ $\therefore \lvert u_n\rvert > \lvert u_{n+1}\rvert$ (2) $\lim_{n\to\infty} \lvert u_n\rvert = \lim_{n\to\infty} \frac{1}{n^2} = 0$ $\therefore$ *By* Leibnitz's test, $\sum u_n$ is convergent. **Answer**
3	Check the convergence of the series $\sum_{n=0}^{\infty}(-1)^n(\sqrt{n+\sqrt{n}}\ -\ \sqrt{n}\ )$ **(Jun-2019-NEW)** **Solution:** The given series is an alternating series $\lvert u_n\rvert = \left\lvert(-1)^n(\sqrt{n+\sqrt{n}} - \sqrt{n})\right\rvert = \sqrt{n+\sqrt{n}} - \sqrt{n} \times \frac{\left(\sqrt{n+\sqrt{n}}+\sqrt{n}\right)}{\left(\sqrt{n+\sqrt{n}}+\sqrt{n}\right)}$ $= \frac{n+\sqrt{n}-n}{\left(\sqrt{n+\sqrt{n}}+\sqrt{n}\right)} = \frac{\sqrt{n}}{\sqrt{n+\sqrt{n}}+\sqrt{n}}$ $\therefore \lvert u_{n+1}\rvert = \frac{\sqrt{n+1}}{\sqrt{(n+1)+(\sqrt{n+1})}+\sqrt{n+1}}$ (1) $\lvert u_n\rvert - \lvert u_{n+1}\rvert = \frac{\sqrt{n}}{\sqrt{n+\sqrt{n}}+\sqrt{n}} - \frac{\sqrt{n+1}}{\sqrt{(n+1)+(\sqrt{n+1})}+\sqrt{n+1}}$ $> 0, for\ all\ n \in N$ $\therefore \lvert u_n\rvert > \lvert u_{n+1}\rvert$ (2) $\lim_{n\to\infty} \lvert u_n\rvert = \lim_{n\to\infty} \frac{\sqrt{n}}{\sqrt{n+\sqrt{n}}+\sqrt{n}} = \frac{1}{2} \neq 0$ $\therefore$ *By* Leibnitz's test, $\sum u_n$ Oscillates. **Answer**
4	Discuss the convergence of the following series $\sum \frac{(-1)^{n-1}}{n\sqrt{n}}$ Discuss the convergence of 1-$\frac{1}{2\sqrt{2}} + \frac{1}{3\sqrt{3}} - \frac{1}{4\sqrt{4}} + \cdots$ **Solution:** The given series is an alternating series $u_n = \frac{(-1)^{n-1}}{n\sqrt{n}}$ $\lvert u_n\rvert = \left\lvert\frac{(-1)^{n-1}}{n\sqrt{n}}\right\rvert = \frac{1}{n\sqrt{n}}$

	$\therefore \lvert u_{n+1}\rvert = \frac{1}{(n+1)\sqrt{n+1}}$ (3) $\lvert u_n\rvert - \lvert u_{n+1}\rvert = \frac{1}{n\sqrt{n}} - \frac{1}{(n+1)\sqrt{n+1}} = \frac{(n+1)\sqrt{n+1}-n\sqrt{n}}{(n^2)(n^2+2n+1)} > 0, for\ all\ n \in N$ $\therefore \lvert u_n\rvert > \lvert u_{n+1}\rvert$ (4) $\lim_{n\to\infty} \lvert u_n\rvert = \lim_{n\to\infty} \frac{1}{n\sqrt{n}} = 0$ $\therefore$ *By* Leibnitz's test, $\sum u_n$ is convergent. **Answer**
5	Test the convergence of the following series $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{\log(n+1)}$ **Solution:** The given series is an alternating series $\lvert u_n\rvert = \left\lvert\frac{(-1)^{n+1}}{\log(n+1)}\right\rvert = \frac{1}{\log(n+1)}$ $\therefore \lvert u_{n+1}\rvert = \frac{1}{\log(n+2)}$ (1) $\lvert u_n\rvert - \lvert u_{n+1}\rvert = \frac{1}{\log(n+1)} - \frac{1}{\log(n+2)} = \frac{\log(n+2)-\log(n+1)}{\log(n+1)\log(n+2)} > 0, for\ all\ n \in N$ $\therefore \lvert u_n\rvert > \lvert u_{n+1}\rvert$ (2) $\lim_{n\to\infty} \lvert u_n\rvert = \lim_{n\to\infty} \frac{1}{\log(n+1)} = 0$ $\therefore$ *By* Leibnitz's test, $\sum u_n$ is convergent. **Answer**
6	Show that the series $\frac{1}{1^p} - \frac{1}{2^p} + \frac{1}{3^p} - \frac{1}{4^p} + \ldots$ converges for p > 0 **Solution:** The given series is an alternating series $\lvert u_n\rvert = \left\lvert\frac{(-1)^{n-1}}{n^p}\right\rvert = \frac{1}{n^p}$ $\therefore \lvert u_{n+1}\rvert = \frac{1}{(n+1)^p}$ (1) $\lvert u_n\rvert - \lvert u_{n+1}\rvert = \frac{1}{n^p} - \frac{1}{(n+1)^p} = \frac{(n+1)^p - n^p}{n^p(n+1)^p} > 0, for\ all\ n \in N\ (\because p > 0)$ $\therefore \lvert u_n\rvert > \lvert u_{n+1}\rvert$ (2) $\lim_{n\to\infty} \lvert u_n\rvert = \lim_{n\to\infty} \frac{1}{n^p} = 0\ (\because p > 0)$

	$\therefore$ *By* Leibnitz's test, $\sum u_n$ is convergent $if\ \ p > 0$. **Answer** **NOTE: If $p < 0$, the conditions (1) and (2) of Leibnitz's test is not satisfied.** **Hence, the given series not convergent $if\ \ p < 0$**
7	Test the convergence of the series $\frac{1}{2} - \frac{2}{5} + \frac{3}{10} - \frac{4}{17} + \cdots$ **Solution:** The given series is an alternating series $u_n = \frac{(-1)^{n-1} n}{n^2+1}$ $\lvert u_n \rvert = \left\lvert \frac{(-1)^{n-1} n}{n^2+1} \right\rvert = \frac{n}{n^2+1}$ $\therefore \lvert u_{n+1} \rvert = \frac{n+1}{(n+1)^2+1} = \frac{n+1}{n^2+2n+2}$ (1) $\lvert u_n \rvert - \lvert u_{n+1} \rvert = \frac{n}{n^2+1} - \frac{n+1}{n^2+2n+2} = \frac{n^3+2n^2+2n-n^3-n-n^2-1}{(n^2+1)(n^2+2n+2)}$ $= \frac{n^2+n-1}{(n^2+1)(n^2+2n+2)} > 0, for\ all\ n \in N$ $\therefore \lvert u_n \rvert > \lvert u_{n+1} \rvert$ (2) $\lim_{n\to\infty} \lvert u_n \rvert = \lim_{n\to\infty} \frac{n}{n^2+1} = 0$ $\therefore$ *By* Leibnitz's test, $\sum u_n$ is convergent. **Answer**
8	Check for convergence of the series $\frac{1}{1.2} - \frac{1}{3.4} + \frac{1}{5.6} - \frac{1}{7.8} + \cdots$ **Solution:** $u_n = \frac{(-1)^{n-1}}{(2n-1)(2n)}$ $\lvert u_n \rvert = \left\lvert \frac{(-1)^{n-1}}{(2n-1)(2n)} \right\rvert = \frac{1}{(2n-1)(2n)}$ $\lvert u_{n+1} \rvert = \frac{1}{(2n+1)(2n+2)}$ (1) $\lvert u_n \rvert - \lvert u_{n+1} \rvert = \frac{1}{(2n-1)(2n)} - \frac{1}{(2n+1)(2n+2)} = \frac{(2n+1)(2n+2)-(2n-1)(2n)}{(2n-1)(2n)(2n+1)(2n+2)}$ $> 0, for\ all\ n \in N$ $\therefore \lvert u_n \rvert > \lvert u_{n+1} \rvert$ (2) $\lim_{n\to\infty} \lvert u_n \rvert = \lim_{n\to\infty} \frac{1}{(2n-1)(2n)} = 0$

	$\therefore$ *By* Leibnitz's test, $\sum u_n$ is convergent. **Answer**
	TO CHECK THE ABSOLUTE AND CONDITIONAL CONVERGENCE
9	Discuss the absolute and conditional convergence of $\sum_{n=1}^{\infty} \frac{(-1)^n n}{n^2+1}$ **Solution:** **(1)To check Absolute Convergence** $\lvert u_n \rvert = \left\lvert \frac{(-1)^n n}{n^2+1} \right\rvert = \frac{n}{n^2+1}$ Take $v_n = \frac{n}{n^2} = \frac{1}{n}$ $\lim_{n\to\infty} \frac{\lvert u_n \rvert}{v_n} = \lim_{n\to\infty} \frac{\frac{n}{n^2+1}}{\frac{1}{n}} = \lim_{n\to\infty} \frac{n^2}{n^2+1} = 1, finite\ and\ non-zero$ Hence, by Limit Comparison test, both $\sum \lvert u_n \rvert$ *and* $\sum v_n$ converge or diverge together. But $\sum v_n = \sum \frac{1}{n}$ is divergent as $p = 1$ $\therefore \sum \lvert u_n \rvert$ is also divergent Hence, $\sum u_n$ is not absolutely convergent. **(2)To check Conditional Convergence** $\lvert u_n \rvert = \left\lvert \frac{(-1)^n n}{n^2+1} \right\rvert = \frac{n}{n^2+1}$ $\therefore \lvert u_{n+1} \rvert = \frac{n+1}{(n+1)^2+1} = \frac{n+1}{n^2+2n+2}$ (3) $\lvert u_n \rvert - \lvert u_{n+1} \rvert = \frac{n}{n^2+1} - \frac{n+1}{n^2+2n+2} = \frac{n^3+2n^2+2n-n^3-n-n^2-1}{(n^2+1)(n^2+2n+2)}$ $= \frac{n^2+n-1}{(n^2+1)(n^2+2n+2)} > 0, for\ all\ n \in N$ $\therefore \lvert u_n \rvert > \lvert u_{n+1} \rvert$ (4) $\lim_{n\to\infty} \lvert u_n \rvert = \lim_{n\to\infty} \frac{n}{n^2+1} = 0$ $\therefore$ *By* Leibnitz's test, $\sum u_n$ is convergent. **The series $\sum u_n$ is convergent and $\sum \lvert u_n \rvert$ is divergent** **Hence, the series is Conditionally convergent.** **Answer**
10	Check the absolute and conditional convergence of the series $\sum_{n=1}^{\infty} \frac{(-1)^n}{\sqrt[3]{n}}$.

Solution:

(1)To check Absolute Convergence

$|u_n| = \left|\frac{(-1)^n}{\sqrt[3]{n}}\right| = \frac{1}{\sqrt[3]{n}}$ is a p − series which is divergent as $p = \frac{1}{3} < 1$

$\therefore \sum |u_n|$ is also divergent

Hence, $\sum u_n$ is not absolutely convergent.

(2)To check Conditional Convergence

The given series is an alternating series, so we use Leibnitz's test.

$|u_n| = \left|\frac{(-1)^n}{\sqrt[3]{n}}\right| = \frac{1}{\sqrt[3]{n}}$

$\therefore |u_{n+1}| = \frac{1}{\sqrt[3]{n+1}}$

(1) $|u_n| - |u_{n+1}| = \frac{1}{\sqrt[3]{n}} - \frac{1}{\sqrt[3]{n+1}} = \frac{\sqrt[3]{n+1} - \sqrt[3]{n}}{\sqrt[3]{n}\,\sqrt[3]{n+1}} > 0, for\ all\ n \in N$

$\therefore |u_n| > |u_{n+1}|$

(2) $\lim_{n\to\infty} |u_n| = \lim_{n\to\infty} \frac{1}{\sqrt[3]{n}} = 0$

$\therefore$ By Leibnitz's test, $\sum u_n$ is convergent.

The series $\sum u_n$ is convergent and $\sum |u_n|$ is divergent

Hence, the series is Conditionally convergent. **Answer**

11 Determine absolute or conditional convergence of the series $\sum_{n=1}^{\infty} (-1)^n \cdot \frac{n^2}{n^3+1}$

Solution:

(1)To check Absolute Convergence

$|u_n| == \left|\frac{(-1)^n n^2}{n^3+1}\right| = \frac{n^2}{n^3+1}$

Take $v_n = \frac{1}{n}$

$$\lim_{n\to\infty} \frac{|u_n|}{v_n} = \lim_{n\to\infty} \frac{\frac{n^2}{n^3+1}}{\frac{1}{n}} = \lim_{n\to\infty} \frac{n^3}{n^3+1} = 1, finite\ and\ non-zero$$

Hence, by Limit Comparison test, both $\sum |u_n|$ and $\sum v_n$ converge or diverge together.

But $\sum v_n = \sum \frac{1}{n}$ is divergent as $p = 1$

$\therefore \sum |u_n|$ is also divergent

Hence, $\sum u_n$ is not absolutely convergent.

(2)To check Conditional Convergence

The given series is an alternating series, so we use Leibnitz's test.

$$|u_n| = \left|\frac{(-1)^n n^2}{n^3+1}\right| = \frac{n^2}{n^3+1}$$

$$\therefore |u_{n+1}| = \frac{(n+1)^2}{(n+1)^3+1} = \frac{n^2+2n+1}{(n+1)^3+1}$$

(1) $|u_n| - |u_{n+1}| = \frac{n^2}{n^3+1} - \frac{n^2+2n+1}{(n+1)^3+1} > 0, for\ all\ n \in N$

$\therefore |u_n| > |u_{n+1}|$

(2) $\lim_{n\to\infty} |u_n| = \lim_{n\to\infty} \frac{n^2}{n^3+1} = 0$

$\therefore$ *By* Leibnitz's test, $\sum u_n$ is convergent.

The series $\sum u_n$ is convergent and $\sum |u_n|$ is divergent

Hence, the series is Conditionally convergent. **Answer**

12 Check for absolute/conditional convergence of $\sum_{n=1}^{\infty} \frac{(-1)^n}{\sqrt{n}+\sqrt{n+1}}$

Solution:

(1)To check Absolute Convergence

$$|u_n| = \left|\frac{(-1)^n}{\sqrt{n}+\sqrt{n+1}}\right| = \frac{1}{\sqrt{n}+\sqrt{n+1}}$$

Take $v_n = \frac{1}{\sqrt{n}}$

$$\lim_{n\to\infty} \frac{|u_n|}{v_n} = \lim_{n\to\infty} \frac{\frac{1}{\sqrt{n}+\sqrt{n+1}}}{\frac{1}{\sqrt{n}}} = \lim_{n\to\infty} \frac{\sqrt{n}}{\sqrt{n}+\sqrt{n+1}} = \frac{1}{1+1} = \frac{1}{2}, finite\ and\ non-zero$$

Hence, by Limit Comparison test, both $\sum |u_n|$ *and* $\sum v_n$ converge or diverge together.

But $\sum v_n = \sum \frac{1}{\sqrt{n}}$ is divergent as $p = \frac{1}{2}$

	$\therefore \sum \lvert u_n \rvert$ is also divergent Hence, $\sum u_n$ is not absolutely convergent. **(2)To check Conditional Convergence** $\lvert u_n \rvert = \left\lvert \frac{(-1)^n}{\sqrt{n}+\sqrt{n+1}} \right\rvert = \frac{1}{\sqrt{n}+\sqrt{n+1}}$ $\therefore \lvert u_{n+1} \rvert = \frac{1}{\sqrt{n+1}+\sqrt{n+2}}$ $(1)\lvert u_n \rvert - \lvert u_{n+1} \rvert = \frac{1}{\sqrt{n}+\sqrt{n+1}} - \frac{1}{\sqrt{n+1}+\sqrt{n+2}} = \frac{(\sqrt{n+1}+\sqrt{n+2})-(\sqrt{n}+\sqrt{n+1})}{(\sqrt{n}+\sqrt{n+1})(\sqrt{n+1}+\sqrt{n+2})} > 0, for\ all\ n \in N$ $\therefore \lvert u_n \rvert > \lvert u_{n+1} \rvert$ $(2) \lim_{n\to\infty} \lvert u_n \rvert = \lim_{n\to\infty} \frac{1}{\sqrt{n}+\sqrt{n+1}} = 0$ $\therefore$ *By* Leibnitz's test, $\sum u_n$ is convergent. **The series $\sum u_n$ is convergent and $\sum \lvert u_n \rvert$ is divergent** **Hence, the series is Conditionally convergent.** **Answer**
13	Is the series absolutely convergent or conditionally convergent? $1 - \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{3}} - \frac{1}{\sqrt{4}} + \cdots \infty$ **Solution:** **(1)To check Absolute Convergence** $\lvert u_n \rvert = \left\lvert \frac{(-1)^{n-1}}{\sqrt{n}} \right\rvert = \frac{1}{\sqrt{n}}$ is a p-series which is divergent **as** $p = \frac{1}{2}$ $\therefore \sum \lvert u_n \rvert$ is divergent Hence, $\sum u_n$ is not absolutely convergent. **(2)To check Conditional Convergence** $\lvert u_n \rvert = \left\lvert \frac{(-1)^{n-1}}{\sqrt{n}} \right\rvert = \frac{1}{\sqrt{n}}$ $\lvert u_{n+1} \rvert = \frac{1}{\sqrt{n+1}}$ $(1)\lvert u_n \rvert - \lvert u_{n+1} \rvert = \frac{1}{\sqrt{n}} - \frac{1}{\sqrt{n+1}} = \frac{(\sqrt{n+1})-(\sqrt{n})}{(\sqrt{n})(\sqrt{n+1})} > 0, for\ all\ n \in N$

Exploring Emerging Technologies

	$\therefore \lvert u_n \rvert > \lvert u_{n+1} \rvert$ (2) $\lim_{n\to\infty} \lvert u_n \rvert = \lim_{n\to\infty} \frac{1}{\sqrt{n}} = 0$ $\therefore$ *By* Leibnitz's test, $\sum u_n$ is convergent. **The series $\sum u_n$ is convergent and $\sum \lvert u_n \rvert$ is divergent** **Hence, the series is Conditionally convergent.** **Answer**
14	Is the series $1-\frac{1}{2}+\frac{1}{3}-\frac{1}{4}+\frac{1}{5}-\frac{1}{6}$...convergent? Does it converge absolutely? Give reasons. **Solution:** **(1)To check Absolute Convergence** $\lvert u_n \rvert = \left\lvert \frac{(-1)^{n-1}}{n} \right\rvert = \frac{1}{n}$ is a p-series which is divergent **as $p = 1$** $\therefore \sum \lvert u_n \rvert$ is divergent Hence, $\sum u_n$ is not absolutely convergent. **(2)To check Conditional Convergence** $\lvert u_n \rvert = \left\lvert \frac{(-1)^{n-1}}{n} \right\rvert = \frac{1}{n}$ $\lvert u_{n+1} \rvert = \frac{1}{n+1}$ (1) $\lvert u_n \rvert - \lvert u_{n+1} \rvert = \frac{1}{n} - \frac{1}{n+1} = \frac{(n+1)-(n)}{(n+1)(n)} = \frac{1}{(n+1)(n)} > 0, for\ all\ n \in N$ $\therefore \lvert u_n \rvert > \lvert u_{n+1} \rvert$ (2) $\lim_{n\to\infty} \lvert u_n \rvert = \lim_{n\to\infty} \frac{1}{n} = 0$ $\therefore$ *By* Leibnitz's test, $\sum u_n$ is convergent. **The series $\sum u_n$ is convergent and $\sum \lvert u_n \rvert$ is divergent** **So it is Conditionally Convergent.** **Answer**
	TOPIC:11 POWER SERIES, RADIUS OF CONVERGENCE OF A POWER SERIES • A power series in powers of (x-a) is defined as a series of the form

$$\sum_{n=0}^{\infty} b_n(x-a)^n = b_0 + b_1(x-a) + b_2(x-a)^2 + \cdots$$

Where $b_0, b_1, b_2, \ldots are\ constants$ known as co efficient of the series.

"a" is constant known as centre of the series. x varies around a.

In particular, when $a = 0$

$$\sum_{n=0}^{\infty} b_n x^n = b_0 + b_1 x + b_2 x^2 + \cdots$$

RADIUS OF CONVERGENCE (R)

$$\frac{1}{R} = \lim_{n\to\infty}\left|\frac{u_{n+1}}{u_n}\right|$$

CONVERGENCE OF POWER SERIES:

Let $\sum_{n=0}^{\infty} b_n x^n$ be a power series.

Then exactly one of the following conditions hold.

(1)The series always converges to $x = 0 \rightarrow R = 0$

(2)The series converges for all x. $\rightarrow R = \infty$

(3)There is some positive number R such that

the series converges for $|x| < R$

the series diverges for $|x| > R$

INTERVAL OF CONVERGENCE:

An interval in which a given series is convergent is called the interval of convergence of the power series.

Exploring Emerging Technologies

SHORT QUESTIONS

1 Find the radius of convergence for the series $\sum_{n=1}^{\infty} \frac{x^n}{n+2}$

Solution:

	$u_n = \frac{x^n}{n+2}$ $u_{n+1} = \frac{x^{n+1}}{n+3}$ $\therefore \frac{1}{R} = \lim_{n\to\infty}\left\|\frac{u_{n+1}}{u_n}\right\| = \lim_{n\to\infty}\left\|\frac{x^{n+1}}{n+3}\times\frac{n+2}{x^n}\right\| = \|x\|$ **∴ By Ratio test,** **(1) $\sum u_n$ is convergent if $\|x\| < 1$** **(2) $\sum u_n$ is divergent if $\|x\| > 1$ and** **(3)The test fails if $\|x\| = 1$ $i.e. x = \pm 1$** ∴ Radius of convergence R = 1 **Answer**
2	Test the convergence and Divergence of the following series $\sum_{n=1}^{\infty} n!(x-4)^n$ **Solution:** Here $u_n = n!\ (x-4)^n$ $\therefore\ u_{n+1} = (n+1)!\,(x-4)^{n+1}$ $\lim_{n\to\infty}\frac{u_{n+1}}{u_n} = \lim_{n\to\infty}\frac{(n+1)!\ (x-4)^{n+1}}{n!\ (x-4)^n}$ $= (x-4)\lim_{n\to\infty}(n+1)$ $= (x-4)\times\infty = \infty > 1,$ for all values of x except $x = 4$ ∴ By ratio test given series is divergent for all values of x, except $x = 4$. When $x = 4$, $u_n = 0$ which is constant and hence convergent. **Answer**
3	Find the radius of convergence and interval of convergence of the series $\sum_{n=0}^{\infty}\frac{x^n}{n!}$ **Solution:** If $x \neq 0$, then

	$\frac{1}{R} = \lim_{n\to\infty}\left\|\frac{u_{n+1}}{u_n}\right\| = \lim_{n\to\infty}\left\|\frac{\frac{x^{n+1}}{(n+1)!}}{\frac{x^n}{n!}}\right\|$ $= \lim_{n\to\infty}\left\|\frac{n!}{(n+1)!x}\right\|$ $= \|x\| \lim_{n\to\infty}\frac{1}{n+1}$ $= 0.$ Therefore, by ratio test, the given series converges absolutely for all x. Thus, the given series converges for all x (Absolutely convergence⇒ Convergence) The radius of convergence **R = ∞** **Answer**
4	Determine the interval of convergence for the series $\sum_{n=1}^{\infty}\frac{x^n}{2^n}, x > 0$ **Solution:** Here $u_n = \frac{x^n}{2^n}$ $\therefore\ u_{n+1} = \frac{x^{n+1}}{2^{n+1}}$ $\frac{1}{R} = \lim_{n\to\infty}\frac{u_{n+1}}{u_n} = \lim_{n\to\infty}\left(\frac{x^{n+1}}{2^{n+1}}\right)\times\left(\frac{2^n}{x^n}\right)$ $= \lim_{n\to\infty}\left(\frac{x}{2}\right)$ $= \left(\frac{x}{2}\right)$ ∴ By Ratio test, (1)The series is convergent if $\left(\frac{x}{2}\right) < 1 \Rightarrow x < 2$ (2)and divergent if $\left(\frac{x}{2}\right) > 1 \Rightarrow x > 2.$ (3) The test fails if $\left(\frac{x}{2}\right) = 1 \Rightarrow x = 2$ **When $x = 2$,** $u_n = \frac{2^n}{2^n} = 1 \Rightarrow \sum u_n = 1 + 1 + 1 \ldots$ which is a divergent series. Hence, the series is convergent for $0 < x < 2$ and Interval of Convergence is $0 < x < 2$

5 Discuss the convergence of the series $\sum_{n=1}^{\infty} \frac{(-1)^n}{2n-1} x^{n+1}$

Solution:

Here $u_n = \frac{(-1)^n}{2n-1} x^{n+1}$

$$u_{n+1} = \frac{(-1)^{n+1}}{2n+1} x^{n+2}$$

$$\frac{u_{n+1}}{u_n} = \frac{(-1)^{n+1}}{2n+1} x^{n+2} \times \frac{2n-1}{(-1)^n x^{n+1}} = \left(-\frac{2n-1}{2n+1}\right) x$$

$$\lim_{n\to\infty} \left|\frac{u_{n+1}}{u_n}\right| = \lim_{n\to\infty} \left| -\frac{\left(2-\frac{1}{n}\right)}{\left(2+\frac{1}{n}\right)} x \right| = |x|$$

$\therefore$ By Ratio test,

(1)The series is convergent if $|x| < 1$ $i.e. -1 < x < 1$

(2)and divergent if $|x| > 1$.

(3) The test fails if $|x| = 1 \Rightarrow x = \pm 1$.

Case: 1 For $x = 1$

$u_n = \frac{(-1)^n}{2n-1} = -1 + \frac{1}{3} - \frac{1}{5} + \cdots$ which is an alternating series

(1)$|u_n| - |u_{n+1}| = \frac{1}{2n-1} - \frac{1}{2n+1} > 0$

(2) $\lim_{n\to\infty} |u_n| = \lim_{n\to\infty} \frac{1}{2n-1} = 0$

$\therefore$ By Leibnitz test, The series is convergent

Case: 2 For $x = (-1)$

$u_n = \frac{(-1)^{2n+1}}{2n-1} = -1 - \frac{1}{3} - \frac{1}{5} \ldots \to -\infty$, which is divergent series

$\therefore$ For $(-1) < x \leq 1$, the series is convergent and $for\ |x| > 1$ the series is divergent.

Answer

6 For the series $\sum_{n=1}^{\infty} \frac{(-1)^n (x+2)^n}{n}$ find the series' radius and interval of convergence. For what values of x does the series converge absolutely, conditionally?

Solution:

Here, $u_n = \frac{(-1)^n (x+2)^n}{n}$,

so $u_{n+1} = \frac{(-1)^{n+1}\,(x+2)^{n+1}}{n+1}$.

Therefore,

$$\frac{1}{R} = \lim_{n\to\infty}\left|\frac{u_{n+1}}{u_n}\right| = \lim_{n\to\infty}\left|\frac{(-1)^{n+1}\,(x+2)^{n+1}}{n+1} \times \frac{n}{(-1)^n\,(x+2)^n}\right|$$

$$= \lim_{n\to\infty}\left|\frac{-n(x+2)}{n+1}\right|$$

$$= |x+2|$$

Hence, by ratio test,

(1)The given series converges absolutely for $|x+2| < 1$ (that is, $-3 < x < -1$)

(2)and diverges for $|x+2| > 1$,

which implies that the radius of convergence is R=1

(3)The test fails for $|x+2| = 1 \Rightarrow x+2 = 1 \;\; and \; x+2 = -1$

$\Rightarrow x = (-1) \; and \; x = (-3)$

Case:1 When $x = -3$, the series becomes

$$\sum_{n=1}^{\infty}\frac{(-1)^n\,(-1)^n}{n} = \sum_{n=1}^{\infty}\frac{(-1)^{2n}}{n} = \sum_{n=1}^{\infty}\frac{1}{n},$$

which is a *p*-series with *p*=1. So, it is divergent,

Case:2 When $x = -1$, the series becomes

$$\sum_{n=1}^{\infty}\frac{(-1)^n\,(1)^n}{n} = \sum_{n=1}^{\infty}\frac{(-1)^n}{n} = -1 + \frac{1}{2} - \frac{1}{3} + \frac{1}{4} - \ldots,$$

which is convergent by Leibniz's theorem because

$$1 > \frac{1}{2} > \frac{1}{3} > \cdots \text{ i.e. } |u_n| > |u_{n+1}|$$

$$\text{And } \lim_{n\to\infty}|u_n| = \lim_{n\to\infty}\left|\frac{(-1)^n}{n}\right| = 0.$$

Therefore, for the given series converges.

interval of convergence is $-3 < x \leq -1$

and interval of absolute convergence is $-3 < x < -1$.

Moreover, the given series converges conditionally at $x = -1$. **Answer**

7 For what values of x do the power series $\sum_{n=1}^{\infty}\frac{(-1)^{n-1}x^{2n-1}}{2n-1}$ converge? OR

For the series $\sum_{n=1}^{\infty}(-1)^{n-1}\frac{x^{2n-1}}{2n-1}$, find the radius and interval of convergence.

Solution:

Here $u_n = \frac{(-1)^{n-1}}{2n-1} x^{2n-1}$

$$u_{n+1} = \frac{(-1)^n}{2n+1} x^{2n+1}$$

$$\frac{u_{n+1}}{u_n} = \frac{(-1)^n}{2n+1} x^{2n+1} \times \frac{2n-1}{(-1)^{n-1} x^{2n-1}} = \left(-\frac{2n-1}{2n+1}\right) x^2$$

$$\frac{1}{R} = \lim_{n\to\infty} \left|\frac{u_{n+1}}{u_n}\right| = \lim_{n\to\infty} \left| -\frac{\left(2-\frac{1}{n}\right)}{\left(2+\frac{1}{n}\right)} x^2 \right| = |x^2|$$

$\therefore$ By Ratio test,

(1)The series is convergent if $|x^2| < 1 \;\; i.e. -1 < x < 1$

(2)and divergent if $|x^2| > 1$.

(3) The test fails if $|x^2| = 1 \;\Rightarrow x = \pm 1$

$\therefore$**Radius of Convergence R = 1**

Case: 1 For $x = 1$

$u_n = \frac{(-1)^{n-1}}{2n-1} = 1 - \frac{1}{3} + \frac{1}{5} - \cdots$ which is an alternating series

(1)$|u_n| - |u_{n+1}| = \frac{1}{2n-1} - \frac{1}{2n+1} > 0$

(2) $\lim_{n\to\infty} |u_n| = \lim_{n\to\infty} \frac{1}{2n-1} = 0$

$\therefore$ By Leibnitz test, The series is convergent

Case: 2 For $x = (-1)$

$$u_n = \frac{(-1)^{3n-2}}{2n-1} = -1 + \frac{1}{3} - \frac{1}{5} \ldots \text{ which is an alternating series}$$

(1)$|u_n| - |u_{n+1}| = \frac{1}{2n-1} - \frac{1}{2n+1} > 0$

(2) $\lim_{n\to\infty} |u_n| = \lim_{n\to\infty} \frac{1}{2n-1} = 0$

$\therefore$ By Leibnitz test, The series is convergent

$\therefore$ For $(-1) \le x \le 1$, the series is convergent.

Interval of Convergence [– 1, 1]

Radius of Convergence R = 1

Since $\sum|u_n|$ is convergent , the series is absolutely convergent for $(-1) \le x \le 1$

8 Find the radius of convergence and interval of convergence for the series $\sum_{n=0}^{\infty} \frac{(-3)^n x^n}{\sqrt{n+1}}$

Solution:

Here $u_n = \frac{(-3)^n}{\sqrt{n+1}} x^n$

$$u_{n+1} = \frac{(-3)^{n+1}}{\sqrt{n+2}} x^{n+1}$$

$$\frac{u_{n+1}}{u_n} = \frac{(-3)^{n+1}}{\sqrt{n+2}} x^{n+1} \times \frac{\sqrt{n+1}}{(-3)^n x^n}$$

$$\frac{1}{R} = \lim_{n\to\infty} \left|\frac{u_{n+1}}{u_n}\right| = \lim_{n\to\infty} \left|\frac{(-3)^{n+1}}{\sqrt{n+2}} x^{n+1} \times \frac{\sqrt{n+1}}{(-3)^n x^n}\right| = \left|\frac{1}{3x}\right|$$

$\therefore$ By Ratio test,

(1)The series is convergent if $\left|\frac{1}{3x}\right| < 1 \;\; i.e. |x| > \frac{1}{3} \Rightarrow (-\frac{1}{3}) < x < \frac{1}{3}$

(2)and divergent if $\left|\frac{1}{3x}\right| > 1 \; i.e. |x| < \frac{1}{3}$

(3) The test fails if $\left|\frac{1}{3x}\right| = 1 \;\Rightarrow x = \pm\frac{1}{3}$

$\therefore$**Radius of Convergence R =** $\frac{1}{3}$

Case: 1 For $x = \frac{1}{3}$

$u_n = \frac{(-1)^n}{\sqrt{n+1}} =$ which is an alternating series

(1)$|u_n| - |u_{n+1}| = \frac{1}{\sqrt{n+1}} - \frac{1}{\sqrt{n+2}} > 0$

(2) $\lim_{n\to\infty} |u_n| = \lim_{n\to\infty} \frac{1}{\sqrt{n+1}} = 0$

$\therefore$ **By Leibnitz test, The series is convergent**

Case: 2 For $x = (-\frac{1}{3})$

$u_n = \frac{1}{\sqrt{n+1}}$

Take $v_n = \frac{1}{\sqrt{n}}$

$\lim_{n\to\infty} \frac{u_n}{v_n} = \lim_{n\to\infty} \frac{\sqrt{n}}{\sqrt{n+1}} = 1$, which is finite and non-zero

Hence, by **Limit Comparison test**, both $\sum u_n$ *and* $\sum v_n$ converge or diverge together.

	But $\sum v_n = \sum \frac{1}{\sqrt{n}}$ **is divergent as** $p = \frac{1}{2} < 1$ $\sum u_n$ is also divergent for $x = (-\frac{1}{3})$ **Hence, the series is convergent for** $x \leq \frac{1}{3}$ **and divergent for** $x > \frac{1}{3}$ $\therefore$ **For** $(-\frac{1}{3}) < x \leq \frac{1}{3}$ **the series is convergent.** **Interval of Convergence** $\left(-\frac{1}{3}, \frac{1}{3}\right]$ **Radius of Convergence R =** $\frac{1}{3}$
9	Find the radius and interval of convergence of the power series $\sum_{n=0}^{\infty} \frac{(-5)^n x^n}{n!}$ **Solution:** Here $u_n = \frac{(-5)^n x^n}{n!}$ $u_{n+1} = \frac{(-5)^{n+1} x^{n+1}}{(n+1)!}$ $\frac{u_{n+1}}{u_n} = \frac{(-5)^{n+1}}{(n+1)!} x^{n+1} \times \frac{n!}{(-5)^n x^n}$ $\frac{1}{R} = \lim_{n\to\infty} \left\lvert \frac{u_{n+1}}{u_n} \right\rvert = \lim_{n\to\infty} \left\lvert \frac{(-5)^{n+1}}{(n+1)!} x^{n+1} \times \frac{n!}{(-5)^n x^n} \right\rvert = \lim_{n\to\infty} \left\lvert \frac{(-5)x}{n+1} \right\rvert = 0$ (1)The series is convergent for all values of x **Radius of Convergence R =** ∞ **Interval of Convergence** $(-\infty, \infty)$ **Answer**
10	Test the convergence of the series $1 - 2x + 3x^2 - 4x^3 + \ldots\ldots, \quad 0 < x < 1$ **Solution:** Here, $u_n = (-1)^{n-1} n x^{n-1}$, So $u_{n+1} = (-1)^n (n+1) x^n$ For $x \neq 0$, $\frac{1}{R} = \lim_{n\to\infty} \left\lvert \frac{u_{n+1}}{u_n} \right\rvert = \lim_{n\to\infty} \left\lvert \frac{(-1)^n (n+1) x^n}{(-1)^{n-1} n x^{n-1}} \right\rvert$ $= \lim_{n\to\infty} \left\lvert (1 + \frac{1}{n}) x \right\rvert$ $= \lvert x \rvert$

	$R = 1$ $\sum\lvert u_n\rvert$ converges if $l = \lvert x\rvert < 1$ i.e. $x < 1$ $(0 < x < 1\ is\ given)$ Thus, the given series converges for $\lvert x\rvert < 1$ and the interval of convergence is $(-1,1)$. **Answer**
11	Find the interval of convergence of the series $1 + 2x + 3x^2 + 4x^3 \ldots$ **Solution:** Here, $u_n = nx^{n-1}$, So $u_{n+1} = (n+1)x^n$. Let us apply ratio test to $\sum u_n$ For $x \neq 0$, $\frac{1}{R} = \lim_{n\to\infty}\left\lvert\frac{u_{n+1}}{u_n}\right\rvert = \lim_{n\to\infty}\left\lvert\frac{(n+1)x^n}{nx^{n-1}}\right\rvert$ $= \lim_{n\to\infty}\left\lvert(1+\frac{1}{n})x\right\rvert$ $= \lvert x\rvert$ $R = 1$ (1) $\sum\lvert u_n\rvert$ converges if $l = \lvert x\rvert < 1$ (2) $\sum\lvert u_n\rvert$ diverges if $l = \lvert x\rvert > 1$ Let us examine the series for $x = \pm 1$ **Case:1** When $x = 1$, the series becomes, $\sum u_n = \sum_{n=1}^{\infty} n = 1 + 2 + 3 + \cdots \to \infty$ Hence, Diverges. **Case:2** When $x = -1$, the series becomes $\sum u_n = \sum_{n=1}^{\infty} n\,(-1)^{n-1} = 1 - 2 + 3 - 4 \ldots$ Therefore, when $x = \pm 1$, the series diverges. Thus, the given series converges for $\lvert x\rvert < 1$ and the interval of convergence is $(-1,1)$**Answer**
12	Test for convergence the series $x - \frac{x^3}{3} + \frac{x^5}{5} + \ldots\ldots, x > 0$ **Solution:** Here $u_n = \frac{(-1)^{n-1}x^{2n-1}}{2n-1}$ $u_{n+1} = \frac{(-1)^n x^{2n+1}}{2n+1}$

$$\therefore \lim_{n\to\infty}\left|\frac{u_{n+1}}{u_n}\right| = \lim_{n\to\infty}\left|\frac{(-1)^n x^{2n+1}}{2n+1}\times\frac{2n-1}{(-1)^{n-1}x^{2n-1}}\right| = x^2$$

$\therefore$ By Ratio test

(1) The series $\sum|u_n|$ is convergent if $x^2 < 1 \Rightarrow x < 1$
(2) The series is divergent if $x^2 > 1$
(3) The test fails if $x^2 = 1 \Rightarrow x = 1$ $(x > 0\ is\ given)$

For $x = 1$

$\sum u_n = \sum\frac{(-1)^{n-1}1^{2n-1}}{2n-1} = \sum\frac{(-1)^{n-1}}{2n-1}$… which is an alternating series

(1)$|u_n| - |u_{n+1}| = \frac{1}{2n-1} - \frac{1}{2n+1} > 0$

(2) $\lim_{n\to\infty}|u_n| = \lim_{n\to\infty}\frac{1}{2n-1} = 0$

$\therefore$ By Leibnitz's test, the series is convergent.

$\therefore$ For $x \le 1$ the series is convergent and for $x > 1$ it is divergent. **Answer**

13 Test for the convergence the series; $\frac{1}{1\cdot2\cdot3}+\frac{x}{4\cdot5\cdot6}+\frac{x^2}{7\cdot8\cdot9}+\ldots\ \ldots$

Solution:

Here, $u_n = \frac{x^{n-1}}{(3n-2)(3n-1)3n}$

$\therefore u_{n+1} = \frac{x^n}{(3n-1)(3n-2)(3n+3)}$

$$\lim_{n\to\infty}\frac{u_{n+1}}{u_n} = \lim_{n\to\infty}\left(\frac{x^n}{(3n-1)(3n-2)(3n+3)}\right)\times\left(\frac{(3n-2)(3n-1)3n}{x^{n-1}}\right) = \lim_{n\to\infty}\frac{x}{\left(1+\frac{1}{n}\right)} = x,$$

$\therefore$By Ratio test,

- $\sum u_n$ converges if $l = x < 1$
- $\sum u_n$ diverges if $l = x > 1$
- The test fails if $l = x = 1$

When $x = 1$,

$$u_n = \frac{1}{(3n-2)(3n-1)3n}$$

Take $v_n = \frac{1}{n^3}$

$$\lim_{n\to\infty}\frac{u_n}{v_n} = \lim_{n\to\infty}\left(\frac{1}{(3n-2)(3n-1)3n}\right)(n^3) = \frac{1}{27}\ ,finite\ and\ non-zero$$

	Hence, by Limit Comparison test, both $\sum u_n$ *and* $\sum v_n$ converge or diverge together. But $\sum v_n = \sum \frac{1}{n^3}$ is convergent as $p = 3 > 1$ $\therefore \sum u_n$ is also convergent for $x = 1$ **Hence, $\sum u_n$ is convergent for $x \leq 1$ And is divergent for $x > 1$** **Answer**

NEW L J INSTITUTE OF ENGINEERING & TECHNOLOGY
LOK JAGRUTI KENDRA
Exploring Emerging Technologies